

EXAMPLE CANDIDATE  
RESPONSE OF EXTERNAL  
SUMMATIVE ASSESSMENT

CHEMISTRY

12  
grade





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### QUESTION 1a (i), (ii), (iii)

For  
Examiner's  
Use

1 This question is about the use of balanced equations.

(a) (i) Copper reacts with concentrated sulfuric acid.

Copper(II) sulfate, sulfur dioxide and water are the products of the reaction.

Construct the balanced equation for this reaction.

..... [2]

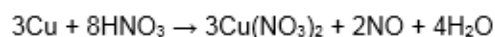
(ii) The mole is used in chemical calculations.

Define the term *mole*.

.....  
..... [1]

(iii) A sample of 192 g of copper reacts with dilute nitric acid.

The equation for the reaction is shown.



Calculate the amount in moles of copper(II) ions / nitrate ions in the copper(II) nitrate formed in this reaction.

Show your working.

amount of copper(II) ion ..... mol

amount of nitrate ion ..... mol [3]

# MARK SCHEME

| Question    | Answer                                                                                                                            | Mark               | Additional guidance                                                                                                                                                                                                       |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 (a) (i)   | $\text{Cu} + 2\text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$                                | 1<br>1<br>[2]      | Correct formulae<br>Balancing – dependent on correct formulae                                                                                                                                                             |
| 1 (a) (ii)  | Amount of substance that contains Avogadro's number of specified particles                                                        | [1]                | <b>Accept</b><br>The relative atomic mass/relative formula mass in grams of any element/compound<br>$6.02 \times 10^{23}$<br>molecules/ions/atoms<br>amount of substance in 12 grams of carbon-12                         |
| 1 (a) (iii) | $192 / 63.5 = 3.0$ moles Cu<br>1:1 (from the equation of reaction)<br>3 moles $\text{Cu}^{2+}$ <b>and</b> 6 moles $\text{NO}_3^-$ | 1<br>1<br>1<br>[3] | <b>Accept</b><br>answer 3.0 - 3.02 moles; 6.0 – 6.02 moles<br><b>ecf</b> for third marking point from moles of copper<br>e.g. if moles of Cu=2 then answer should be 2 moles $\text{Cu}^{2+}$ and 4 moles $\text{NO}_3^-$ |

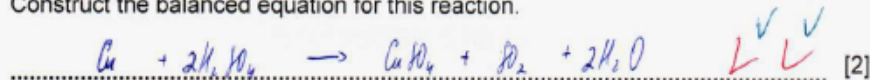
## SAMPLE CANDIDATE RESPONSE GRADED 'A'

1 This question is about the use of balanced equations.

(a) (i) Copper reacts with concentrated sulfuric acid.

Copper(II) sulfate, sulfur dioxide and water are the products of the reaction.

Construct the balanced equation for this reaction.



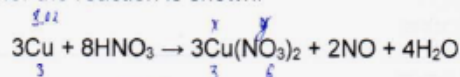
(ii) The mole is used in chemical calculations.

Define the term *mole*.

The number of particles of the substance which is the same  
or the number of atoms in exactly 12 g of C-12 [1]

(iii) A sample of 192 g of copper reacts with dilute nitric acid.

The equation for the reaction is shown.



Calculate the amount in moles of copper(II) ions / nitrate ions in the copper(II) nitrate formed in this reaction.

Show your working.

$$n(\text{Cu}) = \frac{192}{63.5} \approx 3.02 \text{ mol.}$$

$$n(\text{Cu}^{2+}) = 3.02 \text{ mol.}$$

$$n(\text{NO}_3^-) = 6.04 \text{ mol.}$$

$$\frac{3.02}{3} = \frac{x}{3} \quad x = n(\text{Cu}^{2+})$$

$$x = 3.02$$

$$\frac{3.02}{3} = \frac{y}{6} \quad y = n(\text{NO}_3^-)$$

$$y = 6.04$$

amount of copper(II) ion 3.02 mol

amount of nitrate ion 6.04 mol [3]

### EXAMINER'S COMMENTS

This candidate received all six marks. In part 1a(i) the candidate constructed the equation using correct formulae of all compounds reacted and formed during reaction and balanced it. In question 1a(ii) the given definition of the term 'mole' included the exact key idea, and thus was awarded full marks. The answer given in 1a(iii) received three marks for the calculations of the number of moles of copper, copper ions and nitrate ions using correct ratios and formulae.

## SAMPLE CANDIDATE RESPONSE GRADED 'B'

1 This question is about the use of balanced equations.

(a) (i) Copper reacts with concentrated sulfuric acid.

Copper(II) sulfate, sulfur dioxide and water are the products of the reaction.

Construct the balanced equation for this reaction.



(ii) The mole is used in chemical calculations.

Define the term *mole*.

- measure of the amount of compound x [1]

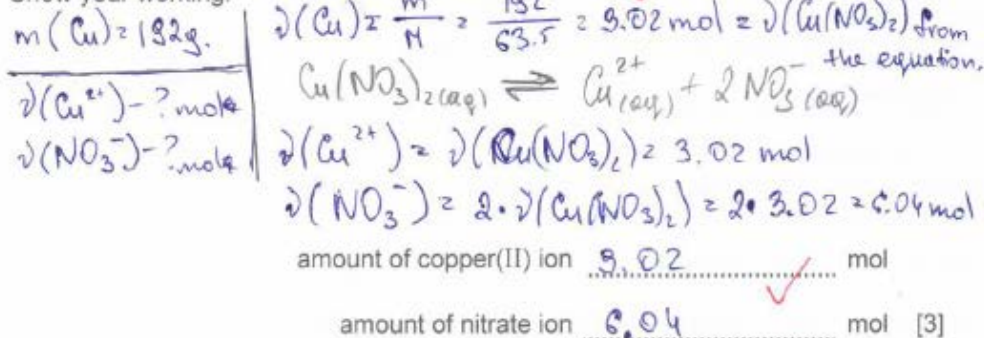
(iii) A sample of 192g of copper reacts with dilute nitric acid.

The equation for the reaction is shown.



Calculate the amount in moles of copper(II) ions / nitrate ions in the copper(II) nitrate formed in this reaction.

Show your working.



### EXAMINER'S COMMENTS

This candidate received five out of six marks. The candidate constructed the equation using correct formulae of all compounds reacted and formed during reaction and balanced it. The given definition was general and did not contain the specific key ideas that could be accepted. The answer in 1a(iii) was awarded full marks for the calculations of the number of moles of copper, copper ions and nitrate ions using correct ratios and formulae.

## SAMPLE CANDIDATE RESPONSE GRADED 'E'

1 This question is about the use of balanced equations.

(a) (i) Copper reacts with concentrated sulfuric acid.

Copper(II) sulfate, sulfur dioxide and water are the products of the reaction.

Construct the balanced equation for this reaction.



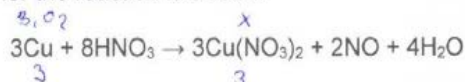
(ii) The mole is used in chemical calculations.

Define the term *mole*.

mole is the quantity to measure the amount of a substance / compound. [1] X

(iii) A sample of 192 g of copper reacts with dilute nitric acid.

The equation for the reaction is shown.



Calculate the amount in moles of copper(II) ions / nitrate ions in the copper(II) nitrate formed in this reaction.

Show your working.

$$n(\text{Cu}) = \frac{192}{63,5} = 3,02 \text{ mol} \quad \checkmark$$

$$x = 3,02 \text{ mol } (\text{Cu}(\text{NO}_3)_2)$$

$$M(\text{Cu}(\text{NO}_3)_2)$$

$$M(\text{NO}_3) = 14 + 48 = 62$$

$$\text{Cu } (63,5) \quad \frac{62}{63,5} = \frac{x}{y} = 3,02$$

$$\begin{aligned} x + y &= 3,02 & x &= 3,02 - y \\ \frac{x}{y} &= 0,97 & 3,02 - y &= 0,97y \\ 3,02 - 1,97y &= 0,97y & 3,02 &= 2,94y \\ y &= 1,03 & x &= 1,99 \end{aligned}$$

amount of copper(II) ion ..... 1,53 ..... mol X

amount of nitrate ion ..... 1,49 ..... mol [3] X

### EXAMINER'S COMMENTS

This candidate received one out of six marks. The answer in 1a(i) was awarded no marks because it included an incorrect formula of copper and a mistake in balancing. The given definition was incorrect due to absence of key ideas as shown in the mark scheme. The candidate received one mark for correct calculation of the number of moles of copper in 1a(iii).



## QUESTION 1b

(b) The equation for a redox reaction is shown.



Write the ionic equation for this redox reaction indicating state symbols.

[2]

## MARK SCHEME

|       |                                                                                                                                                           |          | Note                                                                                 |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------|
| 1 (b) | $\text{MnO}_4^{2-} + \text{H}_2\text{O} \rightarrow \text{MnO}_2 + \text{MnO}_4^- + \text{OH}^-$                                                          | 1        | one mark for the correct ionic species in the equation even if $\text{K}^+$ is shown |
|       | $3\text{MnO}_4^{2-}(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow \text{MnO}_2(\text{s}) + 2\text{MnO}_4^-(\text{aq}) + 4\text{OH}^-(\text{aq})$ | 1<br>[2] | one mark for the balanced ionic equation with state symbols                          |

## SAMPLE CANDIDATE RESPONSE GRADED 'A'

(b) The equation for a redox reaction is shown.

$$3\text{K}_2\text{MnO}_4 + 2\text{H}_2\text{O} \rightarrow \text{MnO}_2 + 2\text{KMnO}_4 + 4\text{KOH}$$

Write the ionic equation for this redox reaction indicating state symbols.

$6\text{K}^+(\text{aq}) + 3\text{MnO}_4^{2-}(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 6\text{K}^+(\text{aq}) + 2\text{MnO}_4^-(\text{aq}) + 4\text{OH}^-(\text{aq}) + \text{MnO}_2(\text{s})$  [2]

2 2

## EXAMINER'S COMMENTS

The candidate received full marks for correct species in the ionic balanced equation with state symbols.

## SAMPLE CANDIDATE RESPONSE GRADED 'C'

(b) The equation for a redox reaction is shown.



Write the ionic equation for this redox reaction indicating state symbols.



✓x

1

### EXAMINER'S COMMENTS

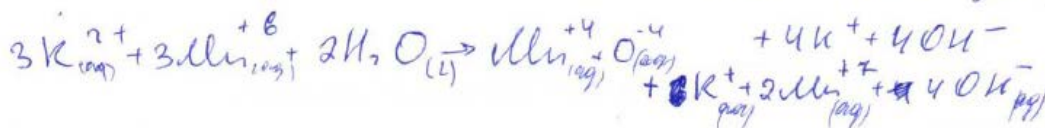
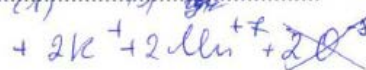
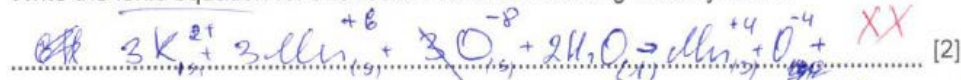
The candidate received partial credit here for the correct balanced ionic equation without state symbols for all included compounds.

## SAMPLE CANDIDATE RESPONSE GRADED 'E'

(b) The equation for a redox reaction is shown.



Write the ionic equation for this redox reaction indicating state symbols.



0

### EXAMINER'S COMMENTS

This candidate received no marks due to an incorrect ionic equation. The candidate lost a mark for states too.

## QUESTION 1c (i), (ii), (iii)

- (c) The enthalpy change of solution of potassium manganate(VII) depends on the lattice energy and the enthalpy change of hydration.

Define the term *enthalpy change of hydration*.

[1]

- (i) Write equations for the enthalpy change of hydration for a potassium ion,  $K^+$ , and a manganate(VII) ion,  $MnO_4^-$ .

Include state symbols in your equations.

enthalpy change of hydration for  $K^+$  \_\_\_\_\_

enthalpy change of hydration for  $MnO_4^-$  \_\_\_\_\_ [2]

- (ii) Draw the energy diagram for the enthalpy change of solution for potassium manganate(VII).

In your diagram, use lattice enthalpy of  $KMnO_4$ , enthalpy of hydration of  $KMnO_4$  and enthalpy of solution of  $KMnO_4$ .

[3]

## MARK SCHEME

|             |                                                                                                    |               |                                                                                                                                                                                                                          |
|-------------|----------------------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 (c) (i)   | (heat changes when) 1 mole of gaseous ions is dissolved to form dilute solution/ in lots of water. | [1]           |                                                                                                                                                                                                                          |
| 1 (c) (ii)  | $K^+(g) \rightarrow K^+(aq)$<br>$MnO_4^-(g) \rightarrow MnO_4^-(aq)$                               | 1<br>1<br>[2] | <b>Accept</b><br>$K^+(g) + aq \rightarrow K^+(aq)$<br>$MnO_4^-(g) + aq \rightarrow MnO_4^-(aq)$                                                                                                                          |
| 1 (c) (iii) |                                                                                                    | [3]           | <b>Note</b><br>1 mark for lattice enthalpy change and correct direction arrow<br>1 mark for change of enthalpy of hydration correct direction arrow<br>1 mark for change of enthalpy of solution correct direction arrow |

## SAMPLE CANDIDATE RESPONSE GRADED 'A'

- (c) The enthalpy change of solution of potassium manganate(VII) depends on the lattice energy and the enthalpy change of hydration.

For  
Examiner  
Use

- (i) Define the term *enthalpy change of hydration*.

Enthalpy change when 1 mole of gaseous ions are dissolved in water under standard conditions. [1]

- (ii) Write equations for the enthalpy change of hydration for a potassium ion,  $K^+$ , and a manganate(VII) ion,  $MnO_4^-$ .

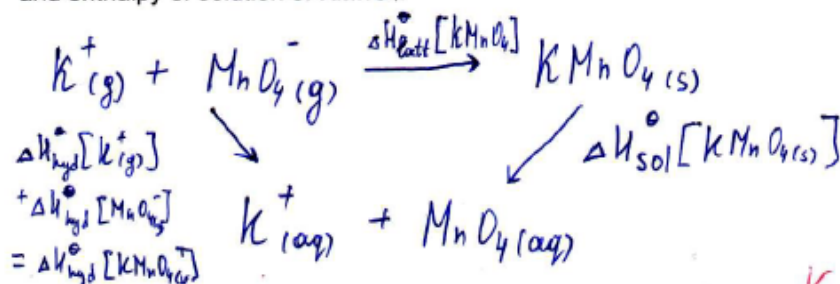
Include state symbols in your equations.

enthalpy change of hydration for  $K^+$   $K^+(g) \rightarrow K^+(aq)$

enthalpy change of hydration for  $MnO_4^-$   $MnO_4^-(g) \rightarrow MnO_4^-(aq)$  [2]

- (iii) Draw the energy diagram for the enthalpy change of solution for potassium manganate(VII).

In your diagram, use lattice enthalpy of  $KMnO_4$ , enthalpy of hydration of  $KMnO_4$  and enthalpy of solution of  $KMnO_4$ .



[3]

### EXAMINER'S COMMENTS

This candidate received all six marks. The given definition of enthalpy change of hydration included the exact key idea and was awarded full marks. The candidate wrote both equations correctly indicating state symbols of all included species in 1c(ii). The given answer in 1c(iii) was awarded three marks for the given diagram that included correct equations required in the task.

## SAMPLE CANDIDATE RESPONSE GRADED 'C'

- (c) The enthalpy change of solution of potassium manganate(VII) depends on the lattice energy and the enthalpy change of hydration.

For  
Examiner's  
Use

- (i) Define the term *enthalpy change of hydration*. —

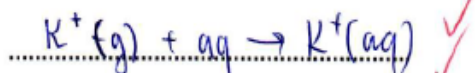
is the enthalpy change when one mole of gaseous ions or is dissolved in a solution. [1]

- (ii) Write equations for the enthalpy change of hydration for a potassium ion,  $K^+$ , and a manganate(VII) ion,  $MnO_4^-$ .

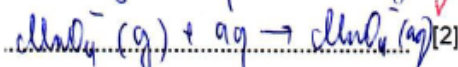
K

Include state symbols in your equations.

enthalpy change of hydration for  $K^+$

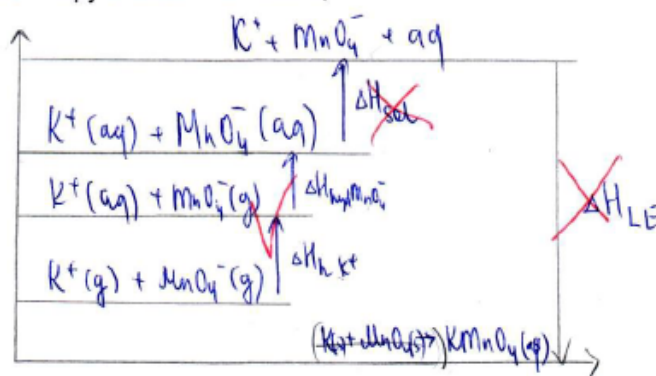


enthalpy change of hydration for  $MnO_4^-$



- (iii) Draw the energy diagram for the enthalpy change of solution for potassium manganate(VII).

In your diagram, use lattice enthalpy of  $KMnO_4$ , enthalpy of hydration of  $KMnO_4$  and enthalpy of solution of  $KMnO_4$ .



[3]

### EXAMINER'S COMMENTS

The candidate received full marks for the correct definition and balanced equations for enthalpy change of hydration of requested ions. However, the answer in 1c (iii) received just one out of three marks because the diagram was composed incorrectly in terms of equations that represented lattice enthalpy and enthalpy of solution of  $KMnO_4$ .

## SAMPLE CANDIDATE RESPONSE GRADED 'E'

(c) The enthalpy change of solution of potassium manganate(VII) depends on the lattice energy and the enthalpy change of hydration.

For  
Examiner's  
Use

(i) Define the term *enthalpy change of hydration*.

when the bond is broken, this process  
require energy, when <sup>there are</sup> bond formation, it release energy [1]

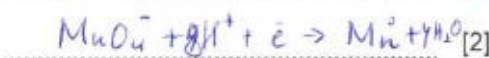
(ii) Write equations for the enthalpy change of hydration for a potassium ion,  $K^+$ , and a manganate(VII) ion,  $MnO_4^-$ .

Include state symbols in your equations.

enthalpy change of hydration for  $K^+$



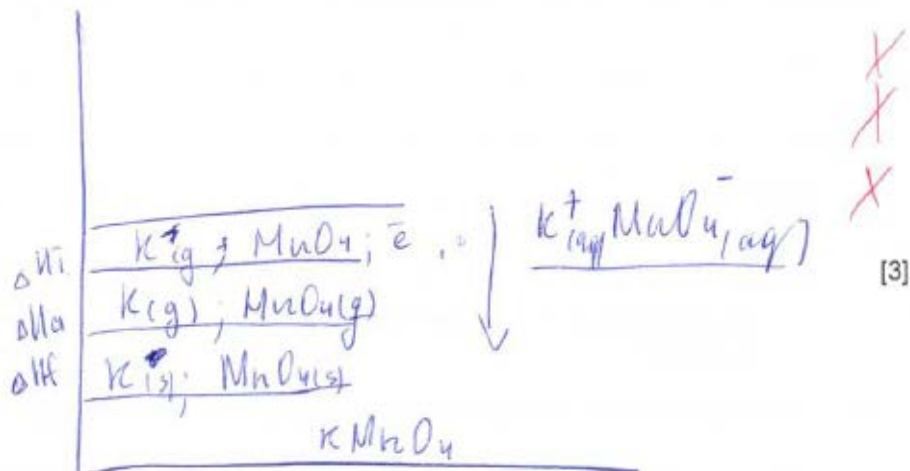
enthalpy change of hydration for  $MnO_4^-$



(iii) Draw the energy diagram for the enthalpy change of solution for potassium manganate(VII).



In your diagram, use lattice enthalpy of  $KMnO_4$ , enthalpy of hydration of  $KMnO_4$  and enthalpy of solution of  $KMnO_4$ .



### EXAMINER'S COMMENTS

The candidate received no marks. The definition was incorrect due to absence of key ideas as shown in the mark scheme. In 1c (ii) the candidate wrote incorrect equations for enthalpy change of hydration for  $K^+$  and  $MnO_4^-$ . The diagram drawn by the candidate was inappropriate because of the arrows and required equations were incorrect.



## QUESTIONS 2a,b (i), (ii)

2 Elemental sulfur is found widespread in nature.

For  
Examiner's  
Use

(a) State one source of sulfur in nature.

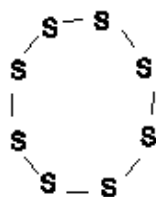
..... [1]

(b) Sulfur has several allotropes.

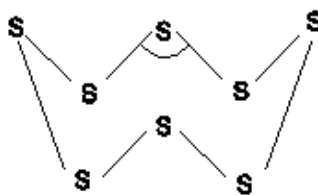
(i) Define the term *allotropes*.

..... [1]  
.....

Two allotropes of sulfur are monoclinic sulfur and rhombic sulfur. Their structures are shown.



rhombic sulphur



monoclinic sulfur

(ii) Identify one similarity and one difference between these two allotropes.

similarity ..... [1]  
.....

difference ..... [1]  
.....

## MARK SCHEME

|            |                                                                                                                                                                                                                                                                                                                                                |                              |                                                                                          |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------|
| 2 (a)      | Sulfur found in nature as brimstone/ /coal/crude oil/volcanoes                                                                                                                                                                                                                                                                                 | [1]                          | <b>Accept</b><br>minerals and salts                                                      |
| 2 (b) (i)  | Being the same element, it has different forms in the same state                                                                                                                                                                                                                                                                               | [1]                          | <b>Accept</b><br><br>Different crystal or molecular lattices of the same element         |
| 2 (b) (ii) | <p><i>Similarity:</i><br/>they are both S<sub>8</sub> / both have eight sulfur atoms in the molecule/ they're both yellow in colour, have single bonds and are in a ring.</p> <p><i>Difference:</i><br/>Monoclinic Sulphur is unstable at temperatures below 96°C/ β-sulfur, non-stable, the puckered ring/ <u>prismatical</u>/ bond angle</p> | <p>1</p> <p>1</p> <p>[2]</p> | <p><b>Accept</b></p> <p>Must have a comment about both rhombic and monoclinic sulfur</p> |

## SAMPLE CANDIDATE RESPONSE GRADED 'A'

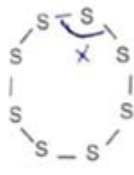
2 Elemental sulfur is found widespread in nature.

(a) State one source of sulfur in nature. *as sulfur oxides S<sub>2</sub>O<sub>3</sub>*  
*non-ferrous metals, monoclinic sulfur, volcanoes after eruption*

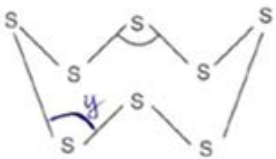
(b) Sulfur has several allotropes.

(i) Define the term *allotropes*.  
*allotropes - different structures of the same element, different physical states of the element* [1]

Two allotropes of sulfur are monoclinic sulfur and rhombic sulfur. Their structures are shown.



rhombic sulphur



monoclinic sulfur

(ii) Identify one similarity and one difference between these two allotropes.

similarity *each sulfur atom has two bonds, bonded to two sulfur atoms* [1]

difference *rhombic sulphur arranged in a oval shaped rhombic way, whereas the monoclinic sulfur has the shape of a chair with smaller bond angle between two bonds.* [1]

*rhombic has higher bond angle between two bonds (∠x > ∠y)*

For Examiner Use



## EXAMINER'S COMMENTS

This response received four out of four possible marks. The candidate was awarded one mark in part (a) for stating correct sulfur source in nature. In part b(i) the candidate defined the term allotropes correctly using key words as shown in the mark scheme. In b(ii) candidate identified similarities and differences between two sulfur allotropes correctly, and thus was awarded full marks.

## SAMPLE CANDIDATE RESPONSE GRADED 'C'

2 Elemental sulfur is found widespread in nature.

(a) State one source of sulfur in nature.

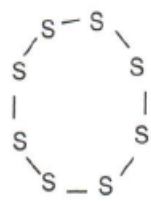
..... Ores of metals ..... [1] X 0

(b) Sulfur has several allotropes.

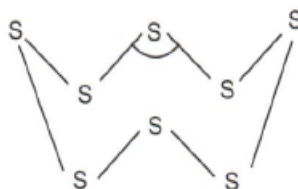
(i) Define the term *allotropes*.

..... allotropes - different shapes/structures  
..... of one atom ..... [1] X 0

Two allotropes of sulfur are monoclinic sulfur and rhombic sulfur. Their structures are shown.



rhombic sulphur



monoclinic sulfur

(ii) Identify one similarity and one difference between these two allotropes.

similarity sulphur's allotropes, same number  
of S and the same bonds between S. [1] ✓ 1

difference monoclinic stable (96-116), rhombic  
(no more than 96), different shapes, properties. [1] 1

For  
Examiner's  
Use

## EXAMINER'S COMMENTS

This response received two out of four possible marks. The candidate lost one mark in part (a) because the given answer was general. In part b(i) the candidate lost one mark, because the given definition was incorrect as instead of words 'same element' the candidate used 'one atom'. Allotropes may contain different types of atoms. In b(ii) the candidate identified similarities and differences between two sulfur allotropes correctly, and thus was awarded all marks.

## SAMPLE CANDIDATE RESPONSE GRADED 'E'

2 Elemental sulfur is found widespread in nature.

(a) State one source of sulfur in nature.

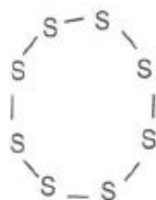
.....stolen in the mountain..... [1]

(b) Sulfur has several allotropes.

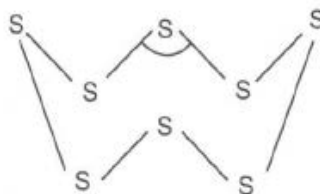
(i) Define the term *allotropes*.

.....Allotropes - is <sup>verse</sup> substance that have different  
.....structure..... [1]

Two allotropes of sulfur are monoclinic sulfur and rhombic sulfur. Their structures are shown.



rhombic sulphur



monoclinic sulfur

(ii) Identify one similarity and one difference between these two allotropes.

similarity .....they produce from one  
.....element - sulfur..... [1]

difference .....the structure: 1) the shape is like  
.....a ball, 2 - crown shape..... [1]

For  
Examin  
Use

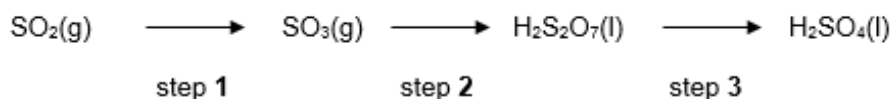
## EXAMINER'S COMMENTS

This response received one out of four possible marks. The candidate lost one mark in part (a) because the given answer was incorrect. In part b(i) the candidate lost one mark because the definition was incorrect as instead of words 'same element' the candidate used 'one substance'. Substance may contain different types of elements. In b(ii) the candidate identified the difference between two sulfur allotropes correctly but was not awarded the mark for identifying the similarity.

### QUESTION 2c (i), (ii), (iii)

- (c) In Kazakhstan sulfuric acid is produced from the waste gases from the extraction of non-ferrous metals.

- (i) The flow chart shows the steps involved in the formation of sulfuric acid.



Write equations for each of the steps. Include state symbols.

step 1 .....

step 2 .....

step 3 ..... [3]

- (ii) Step 1 is an exothermic equilibrium reaction.

The compromise conditions for step 1 are 400 to 450°C, 1 atmosphere pressure and a 1:1 mole ratio of SO<sub>2</sub> to O<sub>2</sub>.

Explain how these three parameters affect the position of the equilibrium and the rate of reaction.

temperature .....

.....

pressure .....

.....

mole ratio .....

..... [3]

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## MARK SCHEME

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                            |                                                                                                                                                                                                                                                                                                       |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 (c) (i) | 1. $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$<br>2. $\text{SO}_3(\text{g}) + \text{H}_2\text{SO}_4(\text{l}) \rightarrow \text{H}_2\text{S}_2\text{O}_7(\text{l})$<br>3. $\text{H}_2\text{S}_2\text{O}_7(\text{l}) + \text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2\text{SO}_4(\text{l})$ / $\text{H}_2\text{S}_2\text{O}_7(\text{l}) + \text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2\text{SO}_4(\text{aq})$ | 1<br>1<br>1<br>[3]         | <b>Do not accept</b><br>unbalanced equations<br><b>Accept</b><br>Reaction arrow in step 1<br><b>2 marks</b> for 3 correct reactions without state symbols<br>Correct multiples including fractions<br>e.g. $\text{SO}_2(\text{g}) + 1/2\text{O}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g})$ |
| 2 (c)(ii) | to increase the rate of reaction without shifting an equilibrium too far to the left<br><br>pressure high enough to have the position of the equilibrium on the right/product<br><br>to shift the position of an equilibrium to the right                                                                                                                                                                                                                                | 1<br><br>1<br><br>1<br>[3] | <b>Accept</b><br><u>owtte</u><br><br><u>owtte</u><br><br><u>owtte</u>                                                                                                                                                                                                                                 |

## SAMPLE CANDIDATE RESPONSE GRADED 'A'

(c) In Kazakhstan sulfuric acid is produced from the waste gases from the extraction of non-ferrous metals.

(i) The flow chart shows the steps involved in the formation of sulfuric acid.



Write equations for each of the steps. Include state symbols.

step 1  $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$  ✓  
 step 2  $\text{SO}_3(\text{g}) + \text{H}_2\text{SO}_4(\text{l}) \rightarrow \text{H}_2\text{S}_2\text{O}_7(\text{l})$  ✓  
 step 3  $\text{H}_2\text{S}_2\text{O}_7(\text{l}) + \text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2\text{SO}_4(\text{aq})$  ✓ [3]

3

(ii) Step 1 is an exothermic equilibrium reaction.

The compromise conditions for step 1 are 400 to 450°C, 1 atmosphere pressure and a 1:1 mole ratio of SO<sub>2</sub> to O<sub>2</sub>.

Explain how these three parameters affect the position of the equilibrium and the rate of reaction.

temperature *as the reaction is exothermic, increasing temperature* *it favours low temperatures.*

pressure *Shifts the equilibrium to the left (reactant) but reaction will be too slow at low temperatures (low rate), 400-450°C - increases rate of reaction but shifts equilibrium to the left, as this comp. would shift the equilibrium to the right into the side with fewer gaseous molecules. In fact, it is not high pressure because at this pressure yield is already 97% (due to T°C) and increasing pressure would be expensive.*

mole ratio *no change on equilibrium* *equilibrium shift to the left right* *rate of reaction increases* *become of more gaseous particles on the left*

*rate of reaction increases as the pressure increase because more collisions happen*

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2

### EXAMINER'S COMMENTS

This response received five out of six possible marks. The candidate was awarded three marks in part c(i) for deducing correct equations with state symbols. In part c(ii) the candidate did not receive full marks as the explanation about compromise temperature related to the rate was correct, but for equilibrium it was not exact according to the mark

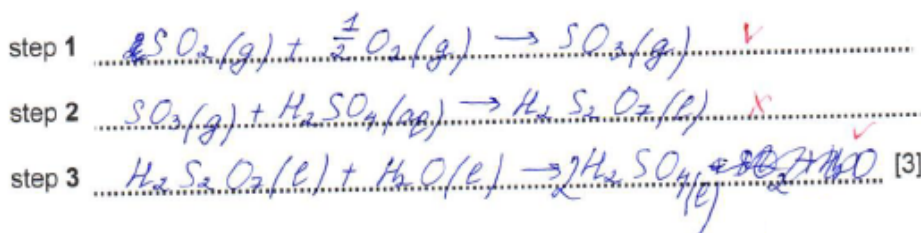
### SAMPLE CANDIDATE RESPONSE GRADED 'C'

(c) In Kazakhstan sulfuric acid is produced from the waste gases from the extraction of non-ferrous metals.

(i) The flow chart shows the steps involved in the formation of sulfuric acid.



Write equations for each of the steps. Include state symbols.





(ii) Step 1 is an exothermic equilibrium reaction.

The compromise conditions for step 1 are 400 to 450°C, 1 atmosphere pressure and a 1:1 mole ratio of SO<sub>2</sub> to O<sub>2</sub>.

Explain how these three parameters affect the position of the equilibrium and the rate of reaction.

temperature t is high enough to maintain fast reaction  
(increases rate when t rises), low enough to maintain good yield (low t favours products)  
pressure high enough for good yield (increasing P favours products), low enough to cut financial costs  
~~increasing P~~ P doesn't affect the rate  
mole ratio maintains the effectiveness of reaction -  
each SO<sub>2</sub> reacts with each O<sub>2</sub>, which gives good yield without many expenses. Mole ratio does not affect the rate of reaction [3]

For  
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Use

11

#### EXAMINER'S COMMENTS

This response received three out of six possible marks. The candidate lost one mark in 2c(i) for the equation at the second stage due to an incorrect state symbol indicated for H<sub>2</sub>SO<sub>4</sub> that should be liquid. In part c(ii) the candidate lost two marks because the explanation about compromise temperature and mole ratio related to the rate or equilibrium was incorrect.

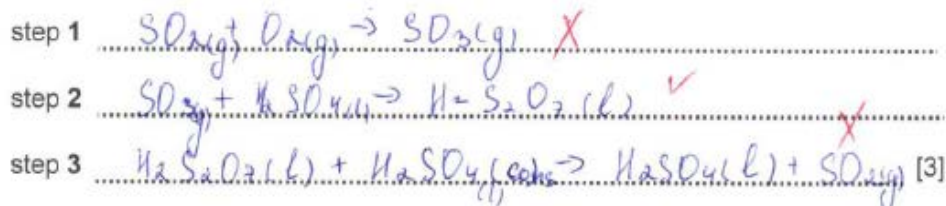
## SAMPLE CANDIDATE RESPONSE GRADED 'E'

(c) In Kazakhstan sulfuric acid is produced from the waste gases from the extraction of non-ferrous metals.

(i) The flow chart shows the steps involved in the formation of sulfuric acid.



Write equations for each of the steps. Include state symbols.



(ii) Step 1 is an exothermic equilibrium reaction.

The compromise conditions for step 1 are 400 to 450°C, 1 atmosphere pressure and a 1:1 mole ratio of SO<sub>2</sub> to O<sub>2</sub>.

Explain how these three parameters affect the position of the equilibrium and the rate of reaction.

temperature  $\text{the rate of reaction increases if temperature increases, equilibrium to the right}$  X

pressure  $\text{if pressure increases, the rate of reaction increases, equilibrium to the left}$  X

mole ratio  $\text{the rate of reaction decreases, if ratio is different, equilibrium shifts to the more moles side}$  X [3]

For  
Examiner  
Use

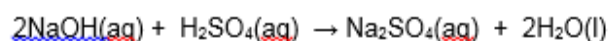
### EXAMINER'S COMMENTS

This response received one out of six possible marks. The candidate lost two marks in 2c(i) for the equations at the first and third stages due to incorrect balancing and incorrect reactants and products. In part c(ii) the candidate received no marks due to incorrect explanation of all compromise conditions related to the rate or / and equilibrium.



## QUESTION 2d

(d) Sodium hydroxide reacts with sulfuric acid as shown in the equation.



A sample of  $300\text{ cm}^3$  of  $1.00\text{ mol dm}^{-3}$   $\text{NaOH}(\text{aq})$  is added to  $100\text{ cm}^3$  of  $1.00\text{ mol dm}^{-3}$   $\text{H}_2\text{SO}_4(\text{aq})$ .

Calculate the pH of the solution formed after the addition. Give your answer to **one** decimal place.

Show your working.

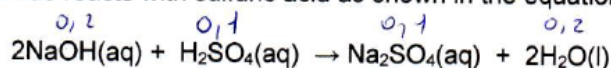
[5]

## MARK SCHEME

|       |                                                                                                                                                                                                                                                                     |                                                                |                                                                                                                                                                                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 (d) | <p>0.10 moles (<math>\text{H}_2\text{SO}_4</math>)</p> <p>0.30 moles (<math>\text{NaOH}</math>)</p> <p>(after the addition) 0.1 mole <math>\text{NaOH}</math></p> <p><math>(0.1/0.4) / 0.25\text{ mol dm}^{-3}</math> <math>\text{NaOH}</math></p> <p>pH = 13.4</p> | <p>1</p> <p>1</p> <p>1</p> <p>1</p> <p>1</p> <p><b>[5]</b></p> | <p><b>Accept</b></p> <p><b>ecf</b> for third marking point</p> <p>e.g.<br/>moles <math>\text{NaOH}</math>-2x moles <math>\text{H}_2\text{SO}_4</math></p> <p><b>ecf</b> for fourth marking point<br/>e.g. moles after addition/0.4</p> <p><b>ecf</b> for fifth marking point<br/>answer 13.0-13.5</p> |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## SAMPLE CANDIDATE RESPONSE GRADED 'A'

(d) Sodium hydroxide reacts with sulfuric acid as shown in the equation.



A sample of 300 cm<sup>3</sup> of 1.00 mol dm<sup>-3</sup> NaOH(aq) is added to 100 cm<sup>3</sup> of 1.00 mol dm<sup>-3</sup> H<sub>2</sub>SO<sub>4</sub>(aq).

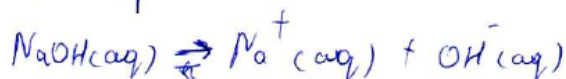
Calculate the pH of the solution formed after the addition. Give your answer to **one** decimal place.

Show your working.

$$n(\text{NaOH}) = 0,3 \text{ dm}^3 \cdot 1 \text{ mol dm}^{-3} = 0,3 \text{ mol} \quad \checkmark$$

$$n(\text{H}_2\text{SO}_4) = 0,1 \text{ dm}^3 \cdot 1 \text{ mol dm}^{-3} = 0,1 \text{ mol} \quad \checkmark$$

H<sub>2</sub>SO<sub>4</sub>(aq) is limiting reagent, it's completely used up.



$$n(\text{OH}^-) = 0,1 \text{ mole} \quad \checkmark \rightarrow [\text{H}^+][\text{OH}^-] = 10^{-14} \quad [\text{OH}^-] = \frac{0,1 \text{ mole}}{0,4} =$$

$$[\text{H}^+] = \frac{K_w}{[\text{OH}^-]} = \frac{10^{-14}}{0,25} = 4 \cdot 10^{-14} \quad [\text{H}^+] = 10^{-13} \text{ mol} \quad 0,25 \frac{\text{mol}}{\text{dm}^3} \quad \checkmark$$

$$\text{pH} = -\log[\text{H}^+] = 13,4 \quad \checkmark$$

[5]

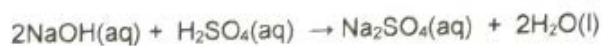
5

### EXAMINER'S COMMENTS

This candidate received all five marks for correct calculation of the amount of NaOH and H<sub>2</sub>SO<sub>4</sub>, final concentration of excess reagent and pH.

## SAMPLE CANDIDATE RESPONSE GRADED 'C'

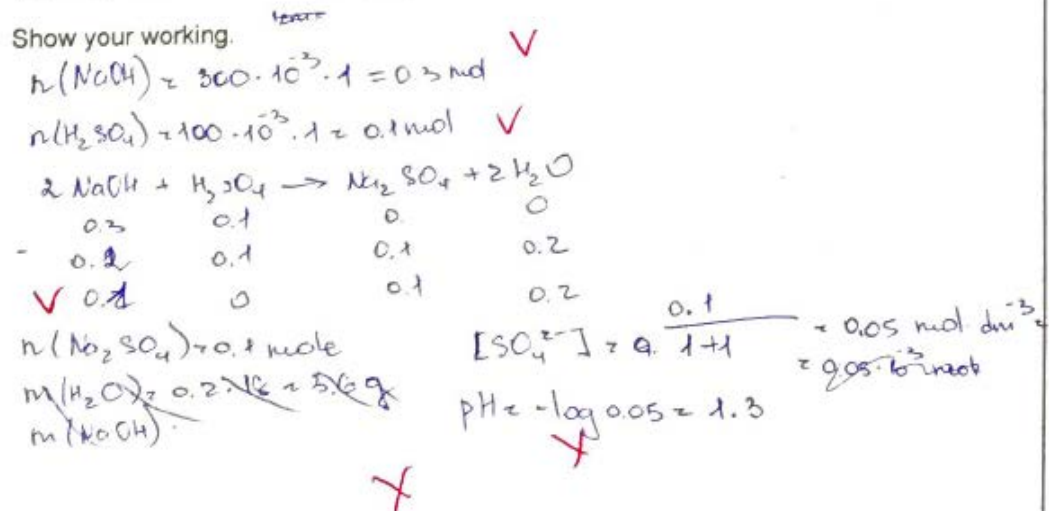
(d) Sodium hydroxide reacts with sulfuric acid as shown in the equation.



A sample of  $300\text{ cm}^3$  of  $1.00\text{ mol dm}^{-3}$   $\text{NaOH}(\text{aq})$  is added to  $100\text{ cm}^3$  of  $1.00\text{ mol dm}^{-3}$   $\text{H}_2\text{SO}_4(\text{aq})$ .

Calculate the pH of the solution formed after the addition. Give your answer to **one** decimal place.

Show your working.



[5]

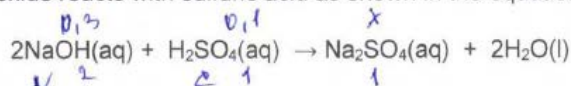
3

### EXAMINER'S COMMENTS

This answer received three out of five possible marks. The candidate lost two marks due to incorrect calculation of the final concentration of excess reagent and pH.

# **SAMPLE CANDIDATE RESPONSE GRADED 'E'**

(d) Sodium hydroxide reacts with sulfuric acid as shown in the equation.



A sample of 300 cm<sup>3</sup> of 1.00 mol dm<sup>-3</sup> NaOH(aq) is added to 100 cm<sup>3</sup> of 1.00 mol dm<sup>-3</sup> H<sub>2</sub>SO<sub>4</sub>(aq).

Calculate the pH of the solution formed after the addition. Give your answer to one decimal place.

Show your working.

Handwritten working:

$\text{pH}(\text{Na}_2\text{SO}_4) = ?$

$V_L = 400 \text{ cm}^3$

$\text{pH} = -\lg[\text{H}^+]$

$[\text{H}^+] = 10^{-\text{pH}}$

$K_a = \frac{[\text{NaOH}][\text{H}_2\text{SO}_4]}{[\text{Na}_2\text{SO}_4]}$

$c = \frac{n}{V}$

$n(\text{NaOH}) = 0,3 \text{ mol} \Rightarrow n(\text{Na}_2\text{SO}_4) = 0,15 \text{ (excess)}$

$n(\text{H}_2\text{SO}_4) = 0,1 \text{ mol} \Rightarrow n(\text{Na}_2\text{SO}_4) = 0,1 \text{ (limit)}$

$n(\text{Na}_2\text{SO}_4) = 0,1$

$V(\text{Na}_2\text{SO}_4) = 400 = 0,4$

$c = \frac{0,1}{0,4} = 0,25 \text{ mol dm}^{-3}$

$\text{pH} = -\lg 0,25 = 0,6$

[5]

2

## **EXAMINER'S COMMENTS**

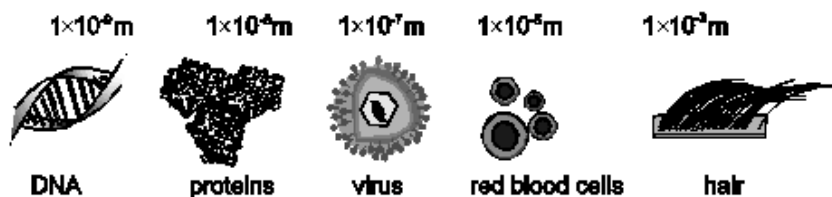
This answer received two out of five possible marks. The candidate lost marks for incorrect determining of excess reagent and calculation of its concentration after reaction and pH of the final solution.

### QUESTION 3a, b (i), (ii)

3 Nanotechnology is a modern area of chemistry.

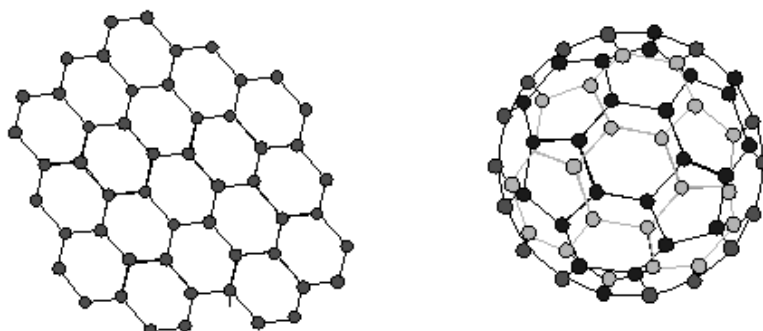
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(a) State which objects in the diagram have the size of a nanoparticle.



[1]

(b) Two nanoparticles of carbon are shown in the diagrams.



(i) Name and describe the structure of one of these nanoparticles.

name .....

description ..... [2]

(ii) Carbon nanoparticles are used in water purification.

Describe one property of carbon nanoparticles which makes them suitable for this.

..... [1]

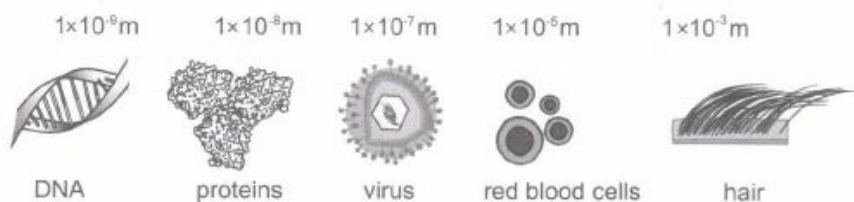
### MARK SCHEME

|            |                                                                                                                                                                                                                                             |                       |                                                                                                                                |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 3 (a)      | Proteins, viruses, DNA                                                                                                                                                                                                                      | [1]                   | 1 mark for <b>all</b> three answer                                                                                             |
| 3 (b) (i)  | Graphene<br><b>OR</b><br>Buckminsterfullerene / fullerene<br><br>Graphene has a hexahedron/hexagonal lattice of one layer carbon atoms.<br><b>OR</b><br>Buckminsterfullerene has a spherical shape with pentagonal and/or hexagonal panels. | 1<br><br>1<br><br>[2] | <b>Accept</b><br><u>buckyball</u><br><br><b>Note</b><br>The name and its description should match for the second marking point |
| 3 (b) (ii) | good Adsorption/large surface area                                                                                                                                                                                                          | 1<br><br>[1]          | <b>Ignore</b><br>small size                                                                                                    |

## SAMPLE CANDIDATE RESPONSE GRADED 'A'

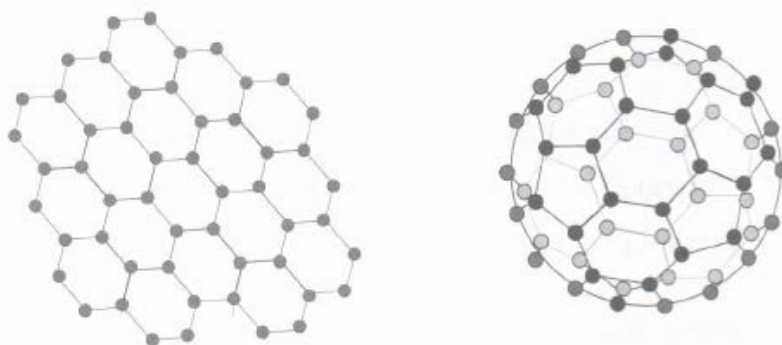
3 Nanotechnology is a modern area of chemistry.

(a) State which objects in the diagram have the size of a nanoparticle.



DNA, proteins, virus X ✓ [1]

(b) Two nanoparticles of carbon are shown in the diagrams.



(i) Name and describe the structure of one of these nanoparticles.

name Fullerene ✓ ✓  
 description C-60, spherical shape ✓ ✓ [2]

(ii) Carbon nanoparticles are used in water purification.

Describe one property of carbon nanoparticles which makes them suitable for this.

They can adsorb unwanted particles ✓ ✓ [1]

### EXAMINER'S COMMENTS

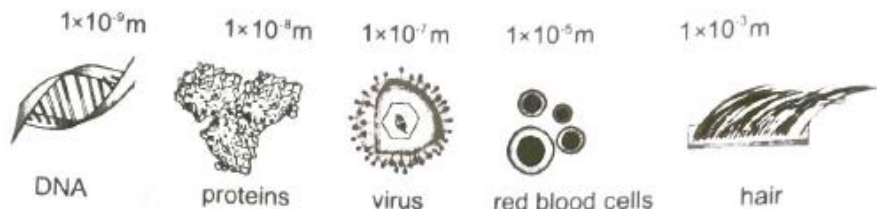
This candidate received all four marks. The candidate distinguished all objects classified as nanoparticles. In part 3b(i) the candidate named and described one of the given nanoparticles correctly. In 3b(ii) the candidate identified one property of carbon nanoparticles that made them suitable for water purification.



## SAMPLE CANDIDATE RESPONSE GRADED 'C'

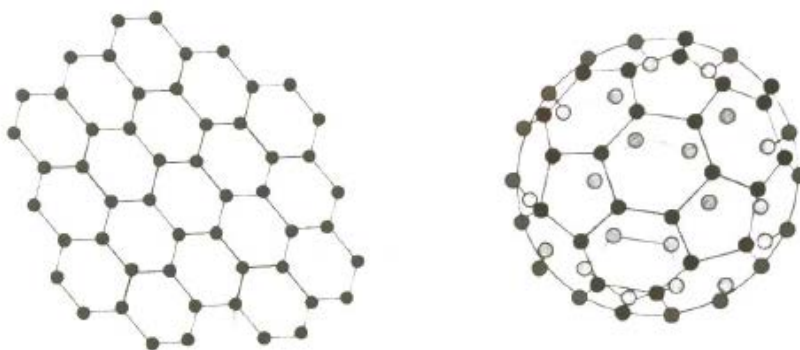
3 Nanotechnology is a modern area of chemistry.

(a) State which objects in the diagram have the size of a nanoparticle.



Virus, DNA, proteins ✓ [1]

(b) Two nanoparticles of carbon are shown in the diagrams.



(i) Name and describe the structure of one of these nanoparticles.

name Graphene ✓  
description One layer of graphite strong for its mass and electronegative [2]

(ii) Carbon nanoparticles are used in water purification.

Describe one property of carbon nanoparticles which makes them suitable for this.  
They can absorb impurities in water. [1]

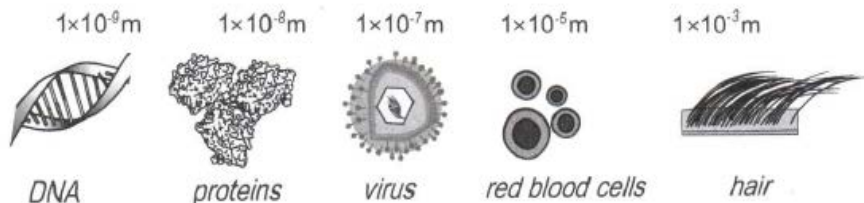
### EXAMINER'S COMMENTS

This candidate received three marks. The candidate distinguished all objects classified as nanoparticles. In part 3b(i) the candidate named and described one of the given nanoparticles correctly. In 3b(ii) instead of the term 'adsorption' the candidate used 'absorption', for this reason he / she lost one mark.

## SAMPLE CANDIDATE RESPONSE GRADED 'E'

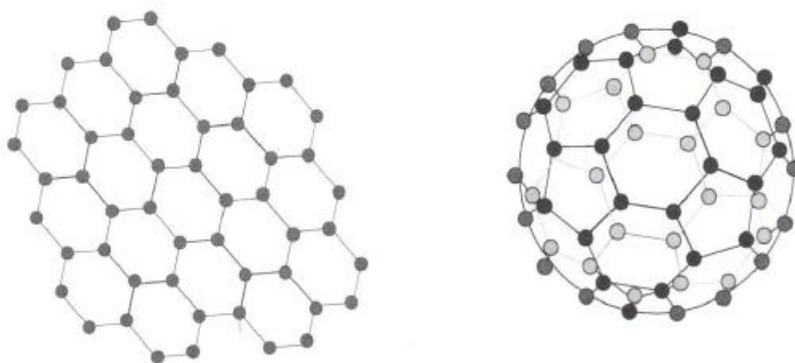
3 Nanotechnology is a modern area of chemistry.

(a) State which objects in the diagram have the size of a nanoparticle.



..... DNA ..... [1]

(b) Two nanoparticles of carbon are shown in the diagrams.



(i) Name and describe the structure of one of these nanoparticles.

name fullerene ✓  
description macromolecular structure X [2]

(ii) Carbon nanoparticles are used in water purification.

Describe one property of carbon nanoparticles which makes them suitable for this.

ability to absorb other particles to own surface X [1]

### EXAMINER'S COMMENTS

This candidate received one out of four marks. The candidate chose just one object, while to receive the mark he / she had to choose all three objects because size of nanoparticles is ranged between 1-100 nm. In part 3b(i) the candidate named the nanoparticle correctly but provided an incorrect description. In 3b(ii) instead of the term 'adsorption' the candidate used 'absorption', for this reason he / she lost one mark.

**QUESTION 3c (i), (ii)**

(c) Nanoparticles and polyethylene glycol (PEG) are used for targeted delivery of drugs.

(i) Describe the process of targeted delivery of drugs into the human body.

You may use a diagram for your answer.

.....

.....

.....

.....

..... [3]

(ii) Describe the advantages **and/or** disadvantages of using nanoparticles and of using PEG for the targeted delivery of drugs into the human body.

.....

.....

.....

..... [3]

For  
Examiner's  
Use

## MARK SCHEME

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                       |                                                                                                |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 3 (c) (i)  | <p>Any <b>three</b> from:</p> <p>An idea that drugs are located inside nanoparticles</p> <p>An idea that polymer chains are bonded with nanoparticles</p> <p>An idea that pores of cell are opened on heating</p> <p>Nanoparticles enter the cell and release drug</p> <p>Nanoparticle then breaks down</p>                                                                                                                                                                                                      | <p>1</p> <p>1</p> <p>1</p> <p>1</p> <p>1</p> <p><b>[max3]</b></p>                                     | <p><b>Accept</b></p> <p>Correct diagram/picture with drugs and polymer</p> <p><u>owtte</u></p> |
| 3 (c) (ii) | <p>Any <b>three</b> from:</p> <p>An idea that less side effects</p> <p>PEG hydrolysis/ transports only two molecules of drug</p> <p>An idea that nanoparticles protect the drugs from digestive system/hydrolysis reaction/ acid</p> <p>Can selectively target cells</p> <p>The molecules can be effectively removed by kidney</p> <p>Small enough to go through the cell membrane</p> <p>Can be made soluble</p> <p>Polymer coating reduces poisonous nature of drugs</p> <p>Nanoparticles may be poisonous</p> | <p>1</p> <p>1</p> <p>1</p> <p>1</p> <p>1</p> <p>1</p> <p>1</p> <p>1</p> <p>1</p> <p><b>[max3]</b></p> | <p><b>Accept</b></p> <p>expensive/high cost</p> <p><b>Accept answer 3 (c) (i)</b></p>          |

## SAMPLE CANDIDATE RESPONSE GRADED 'A'

(c) Nanoparticles and polyethylene glycol (PEG) are used for targeted delivery of drugs.

(i) Describe the process of targeted delivery of drugs into the human body.

You may use a diagram for your answer.

drug molecules are placed inside a nanoparticle with PEG  
and gold coating; when PEG is melted using radiation,  
nanoparticle breaks and drug is released

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Use

3

[3]

(ii) Describe the advantages **and/or** disadvantages of using nanoparticles and of using PEG for the targeted delivery of drugs into the human body.

⊖ <sup>more</sup> expensive than traditional drug delivery

⊕ 1) no need for injection of drugs

2) transferred directly to the active sites of enzymes

3) drugs do not react with hydrochloric acid in stomach

[3]

3

### EXAMINER'S COMMENTS

This candidate received all six marks. In part 3c(i) the candidate described the process of targeted drug delivery correctly using key words. In part c(ii) the candidate described one disadvantage and two advantages of using nanoparticles and PEG in drug delivery. For instance, he / she mentioned 'transferred directly to ...' which means 'can selectively target cells' as written in the mark scheme.

## SAMPLE CANDIDATE RESPONSE GRADED 'C'

(c) Nanoparticles and polyethylene glycol (PEG) are used for targeted delivery of drugs.

(i) Describe the process of targeted delivery of drugs into the human body.

You may use a diagram for your answer.

Drugs which are needed to deliver are placed inside the nanoparticle, for example buckminsterfullerene and transferred to the targeted point <sup>without reaching</sup> where it is released and drug starts functioning as it should.

[3]

(ii) Describe the advantages **and/or** disadvantages of using nanoparticles and of using PEG for the targeted delivery of drugs into the human body.

Advantage: only target point is affected by the drug, safer than using surgery methods or ordinary methods to take drugs.  
Disadvantage: expensive to produce and provide process.

[3]

For  
Examiner's  
Use

2

2

### EXAMINER'S COMMENTS

This candidate received four out of six marks. In part 3c(i) he / she received two marks for writing 'drugs which are needed to deliver are placed inside of nanoparticle ...' and 'transferred to the targeted point ... where it is released ...'. To receive full marks, the answer should include any three ideas according to the mark scheme. In part c(ii) the candidate described one advantage and one disadvantage of using nanoparticles and PEG in drug delivery, which is not enough to receive full marks.



## SAMPLE CANDIDATE RESPONSE GRADED 'E'

(c) Nanoparticles and polyethylene glycol (PEG) are used for targeted delivery of drugs.

(i) Describe the process of targeted delivery of drugs into the human body.

You may use a diagram for your answer.

Drugs are stimulants of function (some body particles, like brain).  
The nanoparticles are macromolecular and they have one or more  
decalized ions which can delivery drugs into the human body.  
DNA is mono particle and in mono-particles there are place for  
drugs can come. [3]

(ii) Describe the advantages **and/or** disadvantages of using nanoparticles and of using PEG for the targeted delivery of drugs into the human body.

health - bad health, attraction to drugs  
- \*enantiomers (one of them not pharmacological active and  
can create a lot of health problems)  
+ economical. lost cost [3]

For  
Examiner  
Use

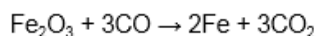
### EXAMINER'S COMMENTS

This candidate received one out of six marks. In part 3c(i) the candidate received no marks due to an incorrect description of the process of targeted delivery of drugs into the human body. In part c(ii) the candidate described one advantage of using nanoparticles and PEG in drug delivery, for this reason he /she received just one mark.

### QUESTION 4a

- 4 Iron is extracted from iron(III) oxide,  $\text{Fe}_2\text{O}_3$ , in a blast furnace.

Both carbon and carbon monoxide reduces iron(III) oxide in a blast furnace.



Some thermochemical data is shown in the table.

|                                                                           | $\text{Fe}_2\text{O}_3(\text{s})$ | $\text{Fe}(\text{s})$ | $\text{C}(\text{s})$ | $\text{CO}(\text{g})$ | $\text{CO}_2(\text{g})$ |
|---------------------------------------------------------------------------|-----------------------------------|-----------------------|----------------------|-----------------------|-------------------------|
| standard enthalpy of formation, $\Delta H_f^\ominus / \text{kJ mol}^{-1}$ | -824.2                            | 0                     | 0                    | -110.5                | -393.5                  |
| standard entropy, $S^\ominus / \text{JK}^{-1} \text{mol}^{-1}$            | 87.4                              | 27.3                  | 5.7                  | 197.6                 | 213.6                   |

- (a) Calculate the standard enthalpy change,  $\Delta H^\ominus$ , and standard entropy change,  $\Delta S^\ominus$ , for the reduction of iron(III) oxide by carbon monoxide.

Show your working.

$$\Delta H^\ominus = \dots\dots\dots \text{kJ mol}^{-1}$$

$$\Delta S^\ominus = \dots\dots\dots \text{JK}^{-1} \text{mol}^{-1} \quad [4]$$

### MARK SCHEME

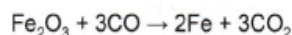
|       |                                                                                                |            |  |
|-------|------------------------------------------------------------------------------------------------|------------|--|
| 4 (a) | For correct use formula/<br>$([3 \times -393.5]) - (-824.2 + [3 \times -110.5])$               | 1          |  |
|       | For correct answer<br>-24.8 ( $\text{kJ} \times \text{mol}^{-1}$ )                             | 1          |  |
|       | For correct use formula/<br>$([2 \times 27.3] + [3 \times 213.6]) - (87.4 + [3 \times 197.6])$ | 1          |  |
|       | For correct answer<br>(+) 15.2 ( $\text{J K}^{-1} \text{mol}^{-1}$ )                           | 1          |  |
|       |                                                                                                | <b>[4]</b> |  |

For  
Examiner's  
Use

## SAMPLE CANDIDATE RESPONSE GRADED 'A'

- 4 Iron is extracted from iron(III) oxide,  $\text{Fe}_2\text{O}_3$ , in a blast furnace.

Both carbon and carbon monoxide reduces iron(III) oxide in a blast furnace.



Some thermochemical data is shown in the table.

|                                                                           | $\text{Fe}_2\text{O}_3(\text{s})$ | $\text{Fe}(\text{s})$ | $\text{C}(\text{s})$ | $\text{CO}(\text{g})$ | $\text{CO}_2(\text{g})$ |
|---------------------------------------------------------------------------|-----------------------------------|-----------------------|----------------------|-----------------------|-------------------------|
| standard enthalpy of formation, $\Delta H_f^\ominus / \text{kJ mol}^{-1}$ | -824.2                            | 0                     | 0                    | -110.5                | -393.5                  |
| standard entropy, $S^\ominus / \text{JK}^{-1} \text{mol}^{-1}$            | 87.4                              | 27.3                  | 5.7                  | 197.6                 | 213.6                   |

- (a) Calculate the standard enthalpy change,  $\Delta H^\ominus$ , and standard entropy change,  $\Delta S^\ominus$ , for the reduction of iron(III) oxide by carbon monoxide.

Show your working.

$$\begin{aligned} & \text{Fe}_2\text{O}_3 + 3\text{CO} \xrightarrow{\Delta H_r^\ominus} 2\text{Fe} + 3\text{CO}_2 \\ & \Delta H_r^\ominus = 2\Delta H_f^\ominus[\text{Fe}(\text{s})] + 3\Delta H_f^\ominus[\text{CO}_2(\text{g})] - \Delta H_f^\ominus[\text{Fe}_2\text{O}_3(\text{s})] - 3\Delta H_f^\ominus[\text{CO}(\text{g})] \\ & \Delta H_r^\ominus = 2 \cdot 0 + 3 \cdot (-393.5) - (-824.2) - 3(-110.5) \\ & \Delta H_r^\ominus = -24.8 \text{ kJ mol}^{-1} \end{aligned}$$

$$\begin{aligned} \Delta S^\ominus &= \Delta S_p^\ominus - \Delta S_r^\ominus \\ \Delta S^\ominus &= 2 \cdot 27.3 + 3 \cdot 213.6 - 87.4 - 3 \cdot 197.6 \\ \Delta S^\ominus &= 15.2 \text{ JK}^{-1} \text{mol}^{-1} \end{aligned}$$

4

For  
Examiner's  
Use

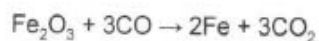
### EXAMINER'S COMMENTS

This candidate received all four marks. The given answer included all calculations that represented correct use of formulae and answers.

# **SAMPLE CANDIDATE RESPONSE GRADED 'C'**

4 Iron is extracted from iron(III) oxide,  $\text{Fe}_2\text{O}_3$ , in a blast furnace.

Both carbon and carbon monoxide reduces iron(III) oxide in a blast furnace.



Some thermochemical data is shown in the table.

|                                                                           | $\text{Fe}_2\text{O}_3(\text{s})$ | $\text{Fe}(\text{s})$ | $\text{C}(\text{s})$ | $\text{CO}(\text{g})$ | $\text{CO}_2(\text{g})$ |
|---------------------------------------------------------------------------|-----------------------------------|-----------------------|----------------------|-----------------------|-------------------------|
| standard enthalpy of formation, $\Delta H_f^\ominus / \text{kJ mol}^{-1}$ | -824.2                            | 0                     | 0                    | -110.5                | -393.5                  |
| standard entropy, $S^\ominus / \text{JK}^{-1} \text{mol}^{-1}$            | 87.4                              | 27.3                  | 5.7                  | 197.6                 | 213.6                   |

- (a) Calculate the standard enthalpy change,  $\Delta H^\ominus$ , and standard entropy change,  $\Delta S^\ominus$ , for the reduction of iron(III) oxide by carbon monoxide.

Show your working.

$$\Delta H_r = \sum \text{bonds broken} - \sum \text{H of bonds formed} = -1155.7 + 1180.5 = 24.8$$

$$\text{bonds broken: } -824.2 - 3 \times 110.5 = -1155.7 \text{ kJ mol}^{-1}$$

$$\text{bonds formed: } -3 \times 393.5 = -1180.5 \text{ kJ mol}^{-1}$$

$$\Delta S_r = \sum S_{\text{pr}} - \sum S_{\text{react}} = 27.3 \times 2 + 3 \times 213.6 - 87.4 - 3 \times 197.6 = 15.2 \text{ JK}^{-1} \text{mol}^{-1}$$

$$\Delta H^\ominus = 24.8 \text{ kJ mol}^{-1}$$

$$\Delta S^\ominus = 15.2 \text{ JK}^{-1} \text{mol}^{-1} \quad [4]$$

For  
Examiner's  
Use

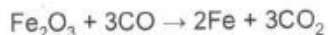
## **EXAMINER'S COMMENTS**

This candidate received two out of four marks. The calculation of standard enthalpy change was incorrect as an incorrect formula was used.

## SAMPLE CANDIDATE RESPONSE GRADED 'E'

- 4 Iron is extracted from iron(III) oxide,  $\text{Fe}_2\text{O}_3$ , in a blast furnace.

Both carbon and carbon monoxide reduces iron(III) oxide in a blast furnace.



Some thermochemical data is shown in the table.

|                                                                         | $\text{Fe}_2\text{O}_3(\text{s})$ | $\text{Fe}(\text{s})$ | $\text{C}(\text{s})$ | $\text{CO}(\text{g})$ | $\text{CO}_2(\text{g})$ |
|-------------------------------------------------------------------------|-----------------------------------|-----------------------|----------------------|-----------------------|-------------------------|
| standard enthalpy of formation, $\Delta H_f^\circ / \text{kJ mol}^{-1}$ | -824.2                            | 0                     | 0                    | -110.5                | -393.5                  |
| standard entropy, $S^\circ / \text{JK}^{-1} \text{mol}^{-1}$            | 87.4                              | 27.3                  | 5.7                  | 197.6                 | 213.6                   |

- (a) Calculate the standard enthalpy change,  $\Delta H^\circ$ , and standard entropy change,  $\Delta S^\circ$ , for the reduction of iron(III) oxide by carbon monoxide.

Show your working.

$$\begin{aligned} \Delta H^\circ &= \sum \Delta H_f^\circ \text{ product} - \sum \Delta H_f^\circ \text{ reactant} \\ \Delta H^\circ &= (-393.5) - (-824.2 - (3 \times 110.5)) = \\ &\quad -393.5 + 1155.5 = 762.0 \text{ kJ mol}^{-1} \\ \Delta S^\circ &= (213.6 + 27.3) - (87.4 + 5.7 + 197.6) = -49.8 \text{ JK}^{-1} \text{mol}^{-1} \\ \Delta H^\circ &= -23.5 \text{ kJ mol}^{-1} \\ \Delta S^\circ &= -49.8 \text{ JK}^{-1} \text{mol}^{-1} \quad [4] \end{aligned}$$

### EXAMINER'S COMMENTS

This candidate received one out of four marks. He / she used the formula for deducing enthalpy change of the reaction correctly, but instead of -393.5, which was given in the table as a standard enthalpy of formation for  $\text{CO}_2$ , in the -393.0 value was used in calculations, for this reason he / she obtained an incorrect answer.

### QUESTION 4b

(b) The equation for the reduction of iron(III) oxide by carbon is shown.



The standard entropy change,  $\Delta S^\ominus$ , for this reaction is  $+542.6 \text{ J K}^{-1} \text{ mol}^{-1}$

Calculate the minimum temperature needed for this reaction to become spontaneous.

Quote your answer to **three** significant figures.

Minimum temperature ..... K [2]

### MARK SCHEME

|       |                                                     |     |  |
|-------|-----------------------------------------------------|-----|--|
| 4 (b) | For correct formula                                 | 1   |  |
|       | For correct answer 908 K with 3 significant figures | 1   |  |
|       |                                                     | [2] |  |

### SAMPLE CANDIDATE RESPONSE GRADED 'A'

(b) The equation for the reduction of iron(III) oxide by carbon is shown.



The standard entropy change,  $\Delta S^\ominus$ , for this reaction is  $+542.6 \text{ J K}^{-1} \text{ mol}^{-1}$

Calculate the minimum temperature needed for this reaction to become spontaneous.

Quote your answer to **three** significant figures.

$\Delta G < 0$  (spont)  $\Delta G = 0$   
 $\Delta G = \Delta H - \Delta S T$   
 $\Delta S T = \Delta H$   $T = \frac{\Delta H}{\Delta S} = \frac{492700 \text{ J mol}^{-1}}{542.6 \text{ J K}^{-1} \text{ mol}^{-1}}$   
 $T = 908 \text{ K}$   
 $\Delta H^\ominus = \Delta H_p - \Delta H_r$   
 $\Delta H^\ominus = 492.7 \text{ kJ mol}^{-1}$

Minimum temperature 908 K ✓ K [2]



### EXAMINER'S COMMENTS

This candidate received full marks.

### SAMPLE CANDIDATE RESPONSE GRADED 'C'

(b) The equation for the reduction of iron(III) oxide by carbon is shown.



The standard entropy change,  $\Delta S^\ominus$ , for this reaction is +542.6 J K<sup>-1</sup> mol<sup>-1</sup>

Calculate the minimum temperature needed for this reaction to become spontaneous.

Quote your answer to three significant figures.

$$\Delta G < 0 \quad \Delta G = \Delta H - T\Delta S < 0 \quad 492.7 < 0.5426T$$
$$T > 908.035 \text{ K} \Rightarrow T_{\min} = 909 \text{ K}$$

Minimum temperature 909 K [2]

### EXAMINER'S COMMENTS

This candidate received one out of two marks. One mark was awarded for the correct formula. The answer 909K was incorrect and the candidate lost one mark.

## SAMPLE CANDIDATE RESPONSE GRADED 'E'

(b) The equation for the reduction of iron(III) oxide by carbon is shown.



The standard entropy change,  $\Delta S^\ominus$ , for this reaction is  $+542.6 \text{ JK}^{-1} \text{ mol}^{-1}$

Calculate the minimum temperature needed for this reaction to become spontaneous.

Quote your answer to **three** significant figures.

$$\begin{array}{r} 492,7 \text{ kJ mol}^{-1} \\ \text{II} \\ 492,7 \text{ kJ mol}^{-1} + 542,6 \frac{\text{J}}{\text{K mol}} = 267,39 \text{ K} \end{array}$$

Minimum temperature 267 K [2]

### EXAMINER'S COMMENTS

This candidate received no marks.

# QUESTION 4c,d (i), (ii)

- (c) The reduction of iron(III) oxide by carbon monoxide is spontaneous at room temperature.

Use ideas about enthalpy change and entropy change to explain the decrease of the minimum temperature in carbon monoxide reduction in comparison to the results calculated in (b).

.....  
 .....  
 .....  
 ..... [2]

- (d) (i) Carbon monoxide reacts quickly with haemoglobin. Describe the function of haemoglobin in the human body.

.....  
 ..... [1]

- (ii) Explain how carbon monoxide prevents haemoglobin to function.

.....  
 .....  
 .....  
 ..... [2]

## MARK SCHEME

|           |                                                                                                                                                                                                                                                           |                              |                                                                                                                                                                      |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 (c)     | $\Delta H^\theta$ for carbon reaction is more than for carbon monoxide/ more endothermic reaction (need high temperature)/reaction for carbon monoxide is more exothermic<br><br>$\Delta S^\theta$ for carbon reaction is larger than for carbon monoxide | 1<br><br>1<br><br><b>[2]</b> | <b>Accept</b><br>as signs > and <<br>Enthalpy contribution in reaction with carbon monoxide is more<br>Entropy contribution in reaction with carbon monoxide is less |
| 4 (d)(i)  | Transport (molecular) oxygen (in blood)                                                                                                                                                                                                                   | <b>[1]</b>                   | <b>owtte</b>                                                                                                                                                         |
| 4 (d)(ii) | idea that (the dative) bond to CO is stronger than that to O <sub>2</sub><br><br>presence of (dative) bond of the CO reduces ability to bond with O <sub>2</sub>                                                                                          | 1<br><br>1<br><br><b>[2]</b> | <b>Accept</b><br>Coordinated bond stronger/polar<br>Need more energy to break between CO and <u>haemoglobin</u>                                                      |

## SAMPLE CANDIDATE RESPONSE GRADED 'A'

- (c) The reduction of iron(III) oxide by carbon monoxide is spontaneous at room temperature.

Use ideas about enthalpy change and entropy change to explain the decrease of the minimum temperature in carbon monoxide reduction in comparison to the results calculated in (b).

$\Delta H^\circ$  of reduction <sup>by</sup> of carbon monoxide is less positive than  $\Delta H^\circ$  of reduction by carbon and  $\Delta S^\circ$  of reduction by carbon monoxide is less positive than the second reaction ✓

[2]

For  
Examiner's  
Use

2

- (d) (i) Carbon monoxide reacts quickly with haemoglobin. Describe the function of haemoglobin in the human body.

Transport of Oxygen / oxygen delivery to respiring cells ✓

[1]

- (ii) Explain how carbon monoxide prevents haemoglobin to function.

CO binds with haemoglobin, forming carboxyhaemoglobin. ✓  
 If CO has a higher affinity to haemoglobin than oxygen, ✓  
 thus it is a competitive inhibitor of <sup>haemoglobin</sup> oxygen. When it <sup>competes with O<sub>2</sub></sup> binds with haemoglobin, oxygen can no longer attach. [2] ✓  
 to haemoglobin, thus it is not transported.  
 Haemoglobin is not functioning properly.

2

### EXAMINER'S COMMENTS

This candidate received all five marks.

## SAMPLE CANDIDATE RESPONSE GRADED 'C'

- (c) The reduction of iron(III) oxide by carbon monoxide is spontaneous at room temperature.

Use ideas about enthalpy change and entropy change to explain the decrease of the minimum temperature in carbon monoxide reduction in comparison to the results calculated in (b).

in the second case (with CO) <sup>more</sup> ~~oxygen~~ gas is produced, which increase level of disorder (entropy change) and enthalpy change of CO<sub>2</sub> <sup>less</sup> than CO → CO<sub>2</sub> ~~is more~~ stable than CO. [2]

- (d) (i) Carbon monoxide reacts quickly with haemoglobin. Describe the function of haemoglobin in the human body.

Haemoglobin binds to oxygen and carrier it to the all body's part. [1]

- (ii) Explain how carbon monoxide prevents haemoglobin to function.

Presents binding with oxygen by reacting with haemoglobin and producing carboxhaemoglobin, which cannot function. Also, CO<sub>2</sub> affects on low percentage of saturation with oxygen in haemoglobin and decrease the pH. [2]

[Total: 11]

For  
Examiner's  
Use

### EXAMINER'S COMMENTS

This candidate received three out of five marks. In part 3c he / she compared the entropy change of two reactions. The second mark was lost due to improper comparison of enthalpy change of given reactions. In task 3d(i) the candidate described the function of haemoglobin correctly. In 4d(ii) the answer clearly represented an idea about reducing the ability of haemoglobin to bind with oxygen. The second mark for this task was not awarded because the candidate failed to explain the exact way by which the carbon monoxide prevented haemoglobin from correct functioning.



## SAMPLE CANDIDATE RESPONSE GRADED 'E'

- (c) The reduction of iron(III) oxide by carbon monoxide is spontaneous at room temperature.

For  
Examiner's  
Use

Use ideas about enthalpy change and entropy change to explain the decrease of the minimum temperature in carbon monoxide reduction in comparison to the results calculated in (b).

Entropy change should be positive to help the spontaneous. Gibbs energy shows feasibility of the reaction and entropy shows spontaneous of reaction. Temperature should be decrease to increase entropy value and spontaneous of reaction, because entropy and temperature directly related decreasing the temperature back increasing of entropy. [2]

0

- (d) (i) Carbon monoxide reacts quickly with haemoglobin. Describe the function of haemoglobin in the human body.

transports red blood cells [1]

- (ii) Explain how carbon monoxide prevents haemoglobin to function.

the O<sub>2</sub> is main component for functioning haemoglobin, if there will be CO the reaction can stop, because there is no favourable conditions. [2]

### EXAMINER'S COMMENTS

This candidate received no marks. In 3c the candidate did not compare the entropy and enthalpy change to explain the minimum temperature at which two reactions occurred, for this reason he / she lost two marks. In 3d(i) the candidate described the function of haemoglobin incorrectly. In 4d(ii) the explanation of the invalid functioning of haemoglobin in presence of carbon monoxide was not specific as required by the mark scheme, for this reason the answer received no marks.



**QUESTION 5a (i), (ii), (iii)**

5 Some protein structures are shown in the diagrams.

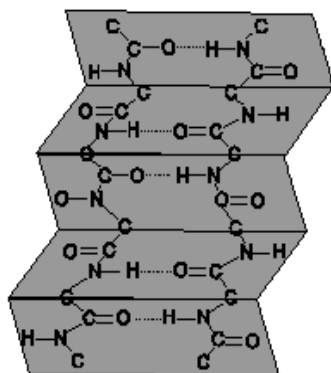
For  
Examiner's  
Use



**G**



**H**



**K**



**L**

(a) (i) Identify the secondary structures shown in G and in K.

G .....

K ..... [2]

(ii) Describe how the structures in G and K are formed.

.....  
 .....  
 .....  
 ..... [2]

(iii) Describe a primary structure and its chemical bonding.

.....  
 .....  
 .....  
 ..... [3]

For  
Examiner's  
Use

## MARK SCHEME

|             |                                                                                                                                               |            |                                |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------------|
| 5 (a) (i)   | <b>G</b> is an alpha helix                                                                                                                    | 1          |                                |
|             | <b>K</b> is a beta pleated sheet / $\beta$ -sheet                                                                                             | 1          |                                |
|             |                                                                                                                                               | <b>[2]</b> |                                |
| 5 (a) (ii)  | Hydrogen bonding                                                                                                                              | 1          |                                |
|             | Between amino acids the H in the N-H bond of the peptide link and the O in the C=O bond of the peptide link                                   | 1          |                                |
|             |                                                                                                                                               | <b>[2]</b> |                                |
| 5 (a) (iii) | The primary structure of a protein is the <u>sequence of amino acid</u> residues in the polypeptide chain that makes up the protein molecule. | 1          | <b>Ignore</b><br>covalent bond |
|             | – NH – CO –                                                                                                                                   | 1          |                                |
|             | Peptide bond / amide linkage between amino acids                                                                                              | 1          |                                |
|             |                                                                                                                                               | <b>[3]</b> |                                |

## SAMPLE CANDIDATE RESPONSE GRADED 'A'

(a) (i) Identify the secondary structures shown in **G** and in **K**.

**G**  $\alpha$ -helix ✓

**K**  $\beta$ -pleated sheet ✓ [2] 2

(ii) Describe how the structures in **G** and **K** are formed.

Hydrogen bonding between ~~peptide~~ carbonyl and N-H bonds in the peptide link between different peptide links of the same chain. ✓

In **G**, the hydrogen bonding is parallel to the axis at a helix. ✓

In **K**, the hydrogen bonding is at right angle ( $90^\circ$ ) of an axis, as  $\beta$ -pleated sheet. [2] 2

(iii) Describe a primary structure and its chemical bonding.

Primary structure is a sequence of amino acids that are bonded to each other via peptide (amide) link. Chemical bonding is covalent (peptide link) between amino acids. There is also ~~side~~ disulphide bridges, which is a covalent bonding. [3] 2

For  
Examiner's  
Use

## EXAMINER'S COMMENTS

This candidate received six out of seven marks. In question 5a(i) he / she named the secondary structures of Q and K correctly and received full marks. The answer for 5a(ii) was complete in terms of explaining how the structures in Q and K were maintained and awarded two marks. The candidate lost one mark in part 5a(iii) for the half description of the primary structure of protein without indicating the functional group of peptide bond.

## SAMPLE CANDIDATE RESPONSE GRADED 'C'

(a) (i) Identify the secondary structures shown in **G** and in **K**.

G α-helix ✓  
K β-pleated sheet ✓ [2]

(ii) Describe how the structures in **G** and **K** are formed.

formed from primary structure of protein,  
hydrogen bonds are formed between the layers  
G is winded form, K is formed into sheet. X  
[2]

(iii) Describe a primary structure and its chemical bonding.

primary structure involves straightforward amino acid sequence ✓  
held by peptide linkage between C=O and N-H as ✓  
well as by covalent bonds throughout the whole chain.  
some amino acids are held due to ester linkage  $-C(=O)-O-C-$  X  
[3]

For  
Examiner's  
Use

## EXAMINER'S COMMENTS

This candidate received five out of seven marks. In question 5a(i) he / she named the secondary structures of Q and K correctly and was awarded full marks. The answer for 5a(ii) was insufficient to receive full marks as despite the candidate named the bond responsible for formation of Q and K structures correctly he / she failed to indicate the functional groups between which these bonds were formed. The candidate lost one mark in 5a(iii) due to contradictory description of the primary structure of protein which contained wrong information about ester linkage.

## SAMPLE CANDIDATE RESPONSE GRADED 'E'

(a) (i) Identify the secondary structures shown in G and in K.

G  $\beta$ -helix X  
 K  ~~$\beta$ -helix~~  $\alpha$ -helix X [2] 0

(ii) Describe how the structures in G and K are formed.

In K is formed by the hydrogen bonds ✓  
 between two polypeptide chains.  
 In G is also hydrogen bonding  
 but spirally shaped structure. + [2] 1

(iii) Describe a primary structure and its chemical bonding.

primary structure has only peptide ✓  
 bond, due to is a weakest structure X  
X  
X [3] 1

For  
Examiner's  
Use

### EXAMINER'S COMMENTS

This candidate received two out of seven marks. In question 5a(i) he / she confused the names of the secondary structures of Q and K and lost all marks. The answer in 5a(ii) was insufficient to receive full marks as despite the candidate named the bond responsible for formation of Q and K structures correctly he / she failed to indicate the functional groups between which these bonds were formed. The candidate lost 2 marks in 5a(iii) due to the contradictory description of the primary structure of protein which did not contain the specific information on which molecules and why were responsible for the formation of peptide bond.

## QUESTION 5 b,c

(b) Define the term *denaturation*.

.....  
 ..... [1]

(c) State **two** different methods of denaturing a protein.

For each method give an example of this type of denaturation.

method 1 .....

example .....

method 2 .....

example ..... [2]

## MARK SCHEME

|       |                                                                                                                                                                                     |     |                                                                                                                                                               |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 (b) | Change of structure/properties of the protein (due to external factors)                                                                                                             | [1] |                                                                                                                                                               |
| 5 (c) | Increase temperature / heating<br><br>cooking of egg / albumin goes solid and/or changes <u>colour</u><br><br>change of pH / add alkali / add acid<br><br>casein in milk goes solid | [2] | <b>Note</b><br>Two correct methods = 1 mark<br>Two correct examples = 1 mark<br>Correct method and example = 1 mark<br><b>Accept</b><br>Any suitable examples |

## SAMPLE CANDIDATE RESPONSE GRADED 'A'

(b) Define the term *denaturation*.

*Denaturation is a reversible process of disruption of 3D-structure of a protein, resulted from adding concentrated alkali/acid; ↑T° or mechanical disruption.* [1]

(c) State **two** different methods of denaturing a protein.

For each method give an example of this type of denaturation.

method 1 *via chemical agent* ✓

example *adding acid or alkali to an egg: white structures are present.* ✓

method 2 *via introducing high temperature.* ✓

example *frying of an egg: albumin starts to condense, harden and whiten.* [2] ✓



### EXAMINER'S COMMENTS

This candidate received all three marks. In 5b the given definition included the main idea '... disruption of 3D-structure...'. In 5c the candidate received full marks for indicating two methods for denaturing protein correctly and providing specific examples for each one.

### SAMPLE CANDIDATE RESPONSE GRADED 'C'

(b) Define the term *denaturation*.

.....the destruction of the structure of the polypeptide resulting.....  
.....in the deformation of the shape of the protein..... [1] ✓ 1

(c) State **two** different methods of denaturing a protein.

For each method give an example of this type of denaturation.

method 1 .....heating over 60°C..... ✓

example .....frying eggs; immune system stops working at higher than 45°C.....

method 2 .....affecting by altering the pH..... 1

example .....hydrolysis with acid or an alkali..... X [2]

### EXAMINER'S COMMENTS

This candidate received two out of three marks. In 5b the given definition included the main key terms according to the mark scheme. In 5c the candidate received only one mark as despite indicating two methods for denaturing protein correctly he / she provided an example for the second method which was too general.



## SAMPLE CANDIDATE RESPONSE GRADED 'E'

(b) Define the term *denaturation*.

~~It is stopping of stop working because of~~  
~~high temperature~~ x [1] 0

(c) State **two** different methods of denaturing a protein.

For each method give an example of this type of denaturation.

method 1 ~~increasing of temperature~~ more than optimal ~~it will denature~~ x  
~~lipase~~  
 example ~~enzyme if we increase more than optimal~~ (66,66)  
 method 2 ~~heavy metal inhibitor~~ x  
 example ~~heavy metal like an inhibitor~~ x [2] 0

### EXAMINER'S COMMENTS

This candidate received no marks. In 5b the given definition was incorrect. In 5c the candidate received no marks as both the given second method and the suggested examples were too general or incorrect.

### QUESTION 5d

(d) A protein contains a disulfide bridge.

Explain how copper(II) ions can change the properties of this protein.

.....  
 .....  
 .....  
 ..... [2]

### MARK SCHEME

|       |                                            |            |                                                                                                                                                                                                                     |
|-------|--------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 (d) | Protein is precipitated                    | 1          | <b>Accept</b><br>If answer is shown as equations<br>$-\text{CH}_2-\text{S}-\text{S}-\text{CH}_2- + \text{Cu}^{2+} \rightarrow 2 (-\text{CH}_2-\text{S}^-) \text{Cu}^{2+}$<br>Solubility of the protein is decreased |
|       | bonding/structure/shape decomposes/changes | 1          |                                                                                                                                                                                                                     |
|       |                                            | <b>[2]</b> |                                                                                                                                                                                                                     |

### SAMPLE CANDIDATE RESPONSE GRADED 'A'

(d) A protein contains a disulfide bridge.

Explain how copper(II) ions can change the properties of this protein.

Copper (II) ions can disrupt the disulfide bridges, ✓  
by binding with sulfur atoms  $-S-Cu-S-$ , ✓  
The covalent bonding between S-S is broken, thus ✓✓  
the ~~st~~ protein starts to unfold, and shape [2]  
changes, leading to denaturation.

2

#### EXAMINER'S COMMENTS

This candidate received full marks. The answer explicitly outlined the mechanism according to which copper (II) ions affected the protein structure and properties.

### SAMPLE CANDIDATE RESPONSE GRADED 'C'

(d) A protein contains a disulfide bridge.

Explain how copper(II) ions can change the properties of this protein.

It can react with this protein and so  
destroy the disulfide bridge, and if it is broken  
then protein structure breaks to ✓ X  
[2]

1

#### EXAMINER'S COMMENTS

This candidate received one out of two marks as he / she indicated the general effect of copper (II) ions on the protein structure but failed to provide a full description of how it was going.

## SAMPLE CANDIDATE RESPONSE GRADED 'E'

(d) A protein contains a disulfide bridge.

Explain how copper(II) ions can change the properties of this protein.

Copper is a heavy metal, due to which it  
bind with substrate, it not give to enzyme  
bind with substrate and decrease its activity

[2]

0

### EXAMINER'S COMMENTS

This candidate received no marks due to incorrect explanation of how copper (II) ions affected the protein structure and properties.

## QUESTION 6 a (i), (ii)

6 (a) Iron can form several complex ions such as  $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$  and  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ .

(i) Write the **full** electronic configuration for  $\text{Fe}^{3+}$ .

$\text{Fe}^{3+}$  ..... [1]

(ii) State and explain which complex ion,  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  or  $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ , shows the greater acidic character.

Complex .....

Explanation .....

.....

..... [2]

## MARK SCHEME

|            |                                                                                                                                                                                                                                                               |                           |                                                                                                                                                                                      |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 (a) (i)  | $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 / [Ar] 3d^5$                                                                                                                                                                                                                   | [1]                       | <b>Ignore</b><br>$4s^0$                                                                                                                                                              |
| 6 (a) (ii) | $[Fe(H_2O)_6]^{3+}$<br>$[Fe(H_2O)_6]^{3+}$ and $Fe^{3+}$ has a greater charge density / smaller ion / higher charge<br>so more <u>polarisation</u> of the O-H bond in the water (ligand) / <u>more</u> distortion of the shared pair of electrons in O-H bond | 0<br>1<br>1<br><b>[2]</b> | <b>Accept</b><br>More polarizing power<br>O-H bond in the water getting weaker produce more $H^+$<br><b>Note</b><br>if $[Fe(H_2O)_6]^{2+}$ given as answer no marks for the question |

## SAMPLE CANDIDATE RESPONSE GRADED 'A'

6 (a) Iron can form several complex ions such as  $[Fe(H_2O)_6]^{3+}$  and  $[Fe(H_2O)_6]^{2+}$ .

(i) Write the **full** electronic configuration for  $Fe^{3+}$ .

$Fe^{3+}$   $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$  [1] ✓

(ii) State **and** explain which complex ion,  $[Fe(H_2O)_6]^{2+}$  or  $[Fe(H_2O)_6]^{3+}$ , shows the greater acidic character.

Complex  $[Fe(H_2O)_6]^{3+}$

Explanation  $[Fe(H_2O)_6]^{3+}$  has a greater charge density, thus the lone pairs on oxygen atom on water molecules are closer. This leads to a high  $\delta^+$  of electron density, and if that is attached to water molecule it has more positive partial charge. It is easier for water molecule in solution to lose  $H^+$  to form  $H_3O^+$  ion. [2] ✓

For  
Examiner's  
Use

## EXAMINER'S COMMENTS

This candidate received all three marks. In 6a(i) the electronic configuration for the ion given in the task was composed correctly. In 6a(ii) the candidate fully reasoned a more acidic character of the complex ion chosen correctly by stating '... greater charge density...', '...the lone pairs on oxygen atom on water are closer'.

## SAMPLE CANDIDATE RESPONSE GRADED 'C'

6 (a) Iron can form several complex ions such as  $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$  and  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ .

(i) Write the **full** electronic configuration for  $\text{Fe}^{3+}$ .

$\text{Fe}^{3+}$  ~~1s~~  $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$  [1] ✓

(ii) State **and** explain which complex ion,  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  or  $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ , shows the greater acidic character.

Complex  $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$

Explanation Because,  $\text{Fe}^{2+}$  is better reducing agent due to its higher lower electronegativity than  $\text{Fe}^{3+}$ . Moreover  $\text{Fe}^{3+}$  has stronger attraction to lone pairs than  $\text{Fe}^{2+}$  [2] ✓

For  
Examiner's  
Use

1

1

### EXAMINER'S COMMENTS

This candidate received full marks. In 6a(i) the electronic configuration for the ion given in the task was composed correctly. In 6a(ii) the candidate lost one mark due to absence of the reasoning of more distortion of bond pair in O-H bond.

## SAMPLE CANDIDATE RESPONSE GRADED 'E'

6 (a) Iron can form several complex ions such as  $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$  and  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ .

(i) Write the **full** electronic configuration for  $\text{Fe}^{3+}$ .  $26 \rightarrow 23$

$\text{Fe}^{3+}$   $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 4p^3$  [1] X

(ii) State **and** explain which complex ion,  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  or  $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ , shows the greater acidic character.

Complex  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$

Explanation strong acid is accepts electrons, strong base donates electrons. therefore  $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$  shows best strong base and  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  is best acid. [2] X X

For  
Examiner's  
Use

0

0



## EXAMINER'S COMMENTS

This candidate received no marks. In 6a(i) the electronic configuration for the given ion was composed incorrectly. In 6a(ii) the answer received no marks due to invalid explanation of the acidic character of the complex ions.

## QUESTIONS 6b,c (i), (ii)

(b) Aqueous carbonate ions react with aqueous  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  in a precipitation reaction.

(i) Construct the balanced equation, including state symbols, for this reaction.

..... [2]

(ii) Describe the observations of this reaction.

..... [1]

(c) Aqueous carbonate ions can also react with aqueous  $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$  in a precipitation reaction.

(i) Construct the balanced equation for this reaction.

..... [1]

(ii) Describe the observations of this reaction.

..... [1]

## MARK SCHEME

|            |                                                                                                                                                                                                                                         |               |                                                                                 |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------------------------------------------------------------------------|
| 6 (b) (i)  | $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{FeCO}_3(\text{s}) + 6\text{H}_2\text{O}(\text{l})$<br>balanced equation<br><br>correct state symbols – dependent on correct formulae | 1<br>1<br>[2] |                                                                                 |
| 6 (b) (ii) | (light) green precipitate                                                                                                                                                                                                               | [1]           | <b>Accept</b><br>Any shades of green                                            |
| 6 (c) (i)  | $2[\text{Fe}(\text{H}_2\text{O})_6]^{3+} + 3\text{CO}_3^{2-} \rightarrow 2[\text{Fe}(\text{H}_2\text{O})_3(\text{OH})_3] + 3\text{CO}_2 + 3\text{H}_2\text{O}$                                                                          | [1]           | <b>Ignore</b><br>State symbols                                                  |
| 6 (c) (ii) | red-brown precipitate/bubbles of gas                                                                                                                                                                                                    | [1]           | <b>Accept</b><br>Brown<br><br><b>Ignore</b><br>Gas formed or carbon dioxide gas |



## SAMPLE CANDIDATE RESPONSE GRADED 'A'

(b) Aqueous carbonate ions react with aqueous  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  in a precipitation reaction.

(i) Construct the balanced equation, including state symbols, for this reaction. ✓✓

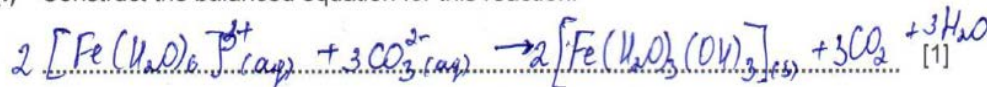


(ii) Describe the observations of this reaction.

green ppt. ✓ [1]

(c) Aqueous carbonate ions can also react with aqueous  $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$  in a precipitation reaction.

(i) Construct the balanced equation for this reaction. ✓



(ii) Describe the observations of this reaction.

bubbles of  $\text{CO}_2$  (effervescence) ✓ [1]

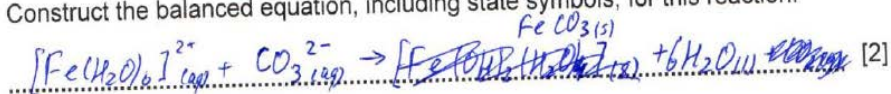
## EXAMINER'S COMMENTS

This candidate received all five marks. The candidate composed the balanced equations correctly with indicating state symbols and provided correct descriptions of the observations of the indicated reactions in 6b and 6c.

## SAMPLE CANDIDATE RESPONSE GRADED 'C'

(b) Aqueous carbonate ions react with aqueous  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  in a precipitation reaction.

(i) Construct the balanced equation, including state symbols, for this reaction. ✓

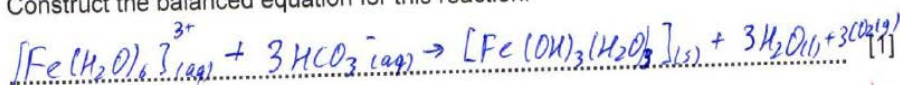


(ii) Describe the observations of this reaction.

White precipitate of  $[\text{Fe}(\text{OH})_3(\text{H}_2\text{O})_3]_{(\text{s})}$  ✓ [1]

(c) Aqueous carbonate ions can also react with aqueous  $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$  in a precipitation reaction.

(i) Construct the balanced equation for this reaction.



(ii) Describe the observations of this reaction.

Brown precipitate  $[\text{Fe}(\text{OH})_3(\text{H}_2\text{O})_3]_{(\text{s})}$  ✓ [1]

## EXAMINER'S COMMENTS

This candidate received three out of five marks. In 6b(i) the balanced equation with state symbols was composed correctly and received full marks. In 6b(ii) the candidate lost one mark because, when describing the observations, instead of the green colour of precipitation he / she mentioned the white colour. In 6c(i) the candidate lost one mark due to incorrect equation of the reaction of carbonate ions with aqueous  $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ . In 6c(ii) the given description of the observations was correct, and thus the candidate received one mark.

## SAMPLE CANDIDATE RESPONSE GRADED 'E'

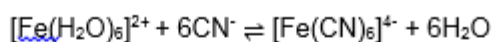
|                                                                                                                              |                                                                                                                                                                                    |     |   |
|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|
| (b) Aqueous carbonate ions react with aqueous $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ in a precipitation reaction.          |                                                                                                                                                                                    |     |   |
| (i)                                                                                                                          | Construct the balanced equation, including state symbols, for this reaction.                                                                                                       |     |   |
|                                                                                                                              | $[\text{Fe}(\text{H}_2\text{O})_6]^{2+} + 6\text{H}_2\text{CO}_3 \rightarrow [\text{Fe}(\text{H}_2\text{O})_4(\text{CO}_3)_2]^{2-} + 2\text{H}_2\text{O} + 4\text{H}_2\text{CO}_3$ | [2] | 0 |
| (ii)                                                                                                                         | Describe the observations of this reaction.                                                                                                                                        |     |   |
|                                                                                                                              | white precipitate is formed                                                                                                                                                        | [1] | 0 |
| (c) Aqueous carbonate ions can also react with aqueous $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ in a precipitation reaction. |                                                                                                                                                                                    |     |   |
| (i)                                                                                                                          | Construct the balanced equation for this reaction.                                                                                                                                 |     |   |
|                                                                                                                              | $[\text{Fe}(\text{H}_2\text{O})_6]^{3+} + 6\text{H}_2\text{CO}_3 \rightarrow [\text{Fe}(\text{H}_2\text{O})_4(\text{CO}_3)_2]^{-} + 2\text{H}_2\text{O} + 4\text{H}_2\text{CO}_3$  | [1] | 0 |
| (ii)                                                                                                                         | Describe the observations of this reaction.                                                                                                                                        |     |   |
|                                                                                                                              | light-blue precipitate                                                                                                                                                             | [1] | 0 |

## EXAMINER'S COMMENTS

This candidate received no marks. He / she composed the balanced equations incorrectly without indicating state symbols in 6b(i) and 6c(i). In 6b(ii) and 6c(ii) the candidate lost marks as he / she described the observations incorrectly.

### QUESTION 6 d, e (i), (ii)

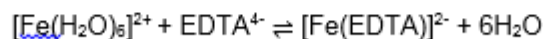
(d) The complex ion  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  reacts with cyanide ions as shown:



Write the expression for the stability constant,  $K_{\text{stab}}$ , of the equilibrium reaction.

..... [1]

(e) The complex ion  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  reacts with  $\text{EDTA}^{4-}$  as shown:



(i) Suggest why  $[\text{Fe}(\text{EDTA})]^{2-}$  is more stable than  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ .

..... [1]

(ii) Use ideas about entropy of the system and Gibbs free energy to suggest why the reaction of  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  with  $\text{EDTA}^{4-}$  is difficult to reverse.

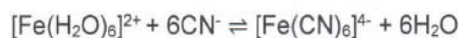
..... [3]

### Mark Scheme

|            |                                                                                                                                                                                                       |                                    |                                                    |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|----------------------------------------------------|
| 6 (d)      | $K_{\text{stab}} = \frac{[\text{Fe}(\text{CN})_6]^{4-}}{[\text{Fe}(\text{H}_2\text{O})_6]^{2+} [\text{CN}^-]^6}$                                                                                      | [1]                                |                                                    |
| 6 (e) (i)  | Chelate effect                                                                                                                                                                                        | [1]                                | <b>Accept</b><br>Description of chelate effect     |
| 6 (e) (ii) | Any <b>three</b> from<br>Entropy change is positive/products have more entropy<br>Because more particles in product<br>Enthalpy change is negligible<br>Gibbs free energy change (large) and negative | 1<br>1<br>1<br>1<br><b>[max 3]</b> | <b>Ignore</b><br><br>Enthalpy change is exothermic |

## SAMPLE CANDIDATE RESPONSE GRADED 'A'

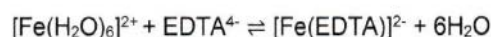
(d) The complex ion  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  reacts with cyanide ions as shown:



Write the expression for the stability constant,  $K_{\text{stab}}$ , of the equilibrium reaction.

$$K_{\text{stab}} = \frac{[\text{Fe}(\text{CN})_6^{4-}]}{[\text{Fe}(\text{H}_2\text{O})_6^{2+}][\text{CN}^-]^6} \quad [1]$$

(e) The complex ion  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  reacts with  $\text{EDTA}^{4-}$  as shown:



(i) Suggest why  $[\text{Fe}(\text{EDTA})]^{2-}$  is more stable than  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ .

$\text{EDTA}^{4-}$  is a polydentate ligand. From entropy perspective, 2 ~~water~~ ions react to make 1 ion and 6 molecules, so it is more favourable - increase in entropy. [1]

(ii) Use ideas about entropy of the system and Gibbs free energy to suggest why the reaction of  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  with  $\text{EDTA}^{4-}$  is difficult to reverse.

In formation of  $[\text{Fe}(\text{EDTA})]^{2-}$ , from 2 ions + that were reacted, 1 ion and 6 water molecules are formed. So there is increase in entropy change, and reaction is more spontaneous ( $\Delta G < 0$ ). [3]

To reverse back, means from 1 ion and 6 molecules produce 2 ions and ~~and~~. It is a decrease in entropy change. ~~The~~ It is a non-spontaneous reaction, and thermodynamically less stable.

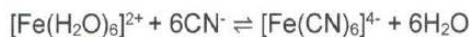
### EXAMINER'S COMMENTS

This candidate received all five marks. In 6d the expression of constant stability was composed correctly. In 6e(i) he / she suggested the correct reason of stability of the complex ion with EDTA ligand. The answer to 6e(ii) included full explanation of the reasons of difficulties related to reversing the reaction outlined in the task with regard to entropy change and Gibbs free energy.



## SAMPLE CANDIDATE RESPONSE GRADED 'C'

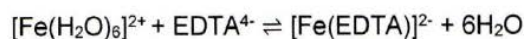
- (d) The complex ion  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  reacts with cyanide ions as shown:



Write the expression for the stability constant,  $K_{\text{stab}}$ , of the equilibrium reaction.

$$K_{\text{stab}} = \frac{[\text{Fe}(\text{CN})_6]^{4-} \cdot [\text{H}_2\text{O}]^6}{[\text{Fe}(\text{H}_2\text{O})_6]^{2+} \cdot [\text{CN}]^6} \quad [1]$$

- (e) The complex ion  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  reacts with  $\text{EDTA}^{4-}$  as shown:



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- (i) Suggest why  $[\text{Fe}(\text{EDTA})]^{2-}$  is more stable than  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ .

entropy increases one  $\text{EDTA}^{4-}$  form six bonds with  $\text{Fe}^{2+}$  while  $\text{H}_2\text{O}$  only one. [1] x 6

- (ii) Use ideas about entropy of the system and Gibbs free energy to suggest why the reaction of  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  with  $\text{EDTA}^{4-}$  is difficult to reverse.

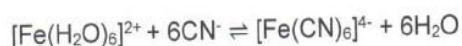
entropy increases when  $[\text{Fe}(\text{EDTA})]^{2-}$  is formed, thus the stability of a complex increases. 2 moles of reactants produce 7 moles of products. It is needed much more energy to break bonds between  $\text{EDTA}^{4-}$  and  $\text{Fe}^{2+}$ . [3] 2

### EXAMINER'S COMMENTS

This candidate received two out of five marks. In 6d the candidate lost one mark for including the concentration of water as well as an incorrect charge of one complex ion in the expression for constant stability. In 6e(i) he / she lost one mark due because the key terms as suggested by the mark scheme were not used. In the 6e(ii) the given explanation was almost full because the candidate reasoned explicitly the difficulty of reversing the reaction with respect to entropy change but not Gibbs free energy change; both were for awarding full marks.

## SAMPLE CANDIDATE RESPONSE GRADED 'E'

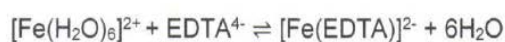
- (d) The complex ion  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  reacts with cyanide ions as shown:



Write the expression for the stability constant,  $K_{\text{stab}}$ , of the equilibrium reaction.

$$K_{\text{stab}} = \frac{[\text{Fe}(\text{CN})_6]^{4-} \cdot [\text{H}_2\text{O}]^6}{[\text{Fe}(\text{H}_2\text{O})_6]^{2+} [\text{CN}^-]^6} \quad [1]$$

- (e) The complex ion  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  reacts with  $\text{EDTA}^{4-}$  as shown:



- (i) Suggest why  $[\text{Fe}(\text{EDTA})]^{2-}$  is more stable than  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ .

$[\text{EDTA}]^{4-}$  is bidentate ligand, while  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  is monodentate ligand. bidentate are have the bond between each other and they more stable.  $\times$  [1]

- (ii) Use ideas about entropy of the system and Gibbs free energy to suggest why the reaction of  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$  with  $\text{EDTA}^{4-}$  is difficult to reverse.

One EDTA molecule replace 6 molecule of water, which means that EDTA molecule's are very big and there is need a lot of the energy and high temperature for reversing this reaction.  $\times$   $\times$  [3]

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## EXAMINER'S COMMENTS

This candidate received one out of five marks. In 6d the candidate lost one mark due to the incorrect expression for constant stability. In the answer to 6e(i), to explain the stability of complex ion with EDTA ligand, the candidate used an incorrect term 'bidentate ligand' instead of 'polydentate ligand'. In 6e(ii) the answer received 1 mark as the candidate described the idea about increasing the number of moles after the reaction but failed to mention about entropy change and Gibbs free energy.



## QUESTIONS 7a (i), (ii), (iii)

7 Alcohols are widely used for some industrial processes.

(a) (i) An alcohol has the molecular formula  $C_4H_{10}O$ .

Draw the structure of three isomers of  $C_4H_{10}O$  that are alcohols.

Name each of the isomers you draw.

[3]

(ii) Alcohols can undergo reactions involving the loss of water.

Name this type of reaction.

[1]

(iii) State the conditions for this reaction.

[1]

## MARK SCHEME

|             |                                                                                                                                                                                                |                                    |                                                                                                                                                                                                  |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7 (a) (i)   | Any <b>three</b> from<br>Butan-1-ol $CH_3-CH_2-CH_2-CH_2OH$<br>Butan-2-ol $CH_3-CH(OH)-CH_2-CH_3$<br>2-methylpropan-1-ol $CH_3-CH(CH_3)-CH_2OH$<br>2-methylpropan-2-ol $CH_3-C(CH_3)(OH)-CH_3$ | 1<br>1<br>1<br>1<br><b>[max 3]</b> | The names to corresponds the structural formulae<br><b>Accept</b><br>$(CH_3)_3COH$<br>without showing the bonds<br>e.g. $CH_3C(CH_3)(OH)CH_3$<br>1 mark for all correct structures without names |
| 7 (a) (ii)  | elimination / dehydration                                                                                                                                                                      | <b>[1]</b>                         |                                                                                                                                                                                                  |
| 7 (a) (iii) | $H^+/H_2SO_4 / H_3PO_4 / Al_2O_3 / ZnCl_2 / SiO_2$<br>Higher than $200^\circ C / 300^\circ C / 400^\circ C$                                                                                    | <b>[1]</b>                         | both answers needed<br><b>Accept</b><br>high temperature<br><b>Ignore</b><br>Heat                                                                                                                |

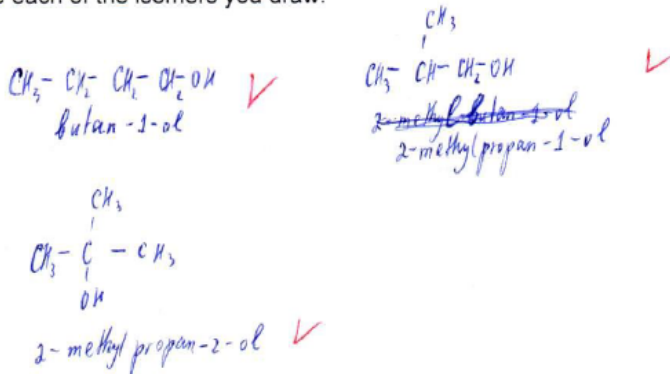
## SAMPLE CANDIDATE RESPONSE GRADED 'A'

7 Alcohols are widely used for some industrial processes.

(a) (i) An alcohol has the molecular formula  $C_4H_{10}O$ .

Draw the structure of three isomers of  $C_4H_{10}O$  that are alcohols.

Name each of the isomers you draw.



[3]

33

(ii) Alcohols can undergo reactions involving the loss of water.

Name this type of reaction.

dehydration ✓ ✓ [1]

11

(iii) State the conditions for this reaction.

$H_2SO_4$  (conc.) and high temperature ✓ ✓ [1]

11

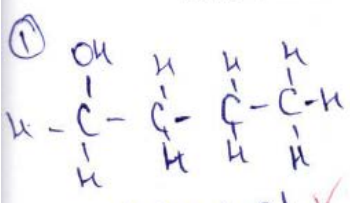
### EXAMINER'S COMMENTS

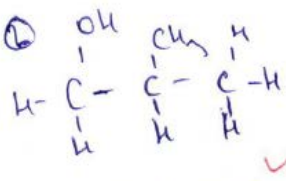
This candidate received all five marks. In 7a(i) the structural formulae of three  $C_4H_{10}O$  isomers were drawn and named correctly. In answers to 7a(ii-iii) the candidate stated the name and conditions for the process described in the task correctly.

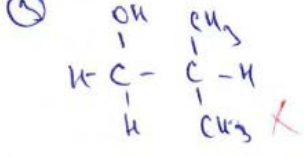
## SAMPLE CANDIDATE RESPONSE GRADED 'C'

7 Alcohols are widely used for some industrial processes.

(a) (i) An alcohol has the molecular formula  $C_4H_{10}O$ .  
 Draw the structure of three isomers of  $C_4H_{10}O$  that are alcohols.  
 Name each of the isomers you draw.

①   
 butan-1-ol ✓

②   
 2-methylpropan-1-ol ✓

③   
 2,2-dimethylethan-1-ol. ✗

[3] 2

(ii) Alcohols can undergo reactions involving the loss of water.  
 Name this type of reaction.  
Dehydration [1] ✓ 1

(iii) State the conditions for this reaction.  
High temperature, high pressure [1] ✗ 0

### EXAMINER'S COMMENTS

This candidate received three out of five marks. In 7a(i) he / she lost one mark due to an incorrect name of the third drawn structural isomer. The answer to 7a(ii) received one mark for the correct name of the process described in the task, but the candidate also lost one mark in 7a(ii) due to failure to indicate all conditions as required by the mark scheme.

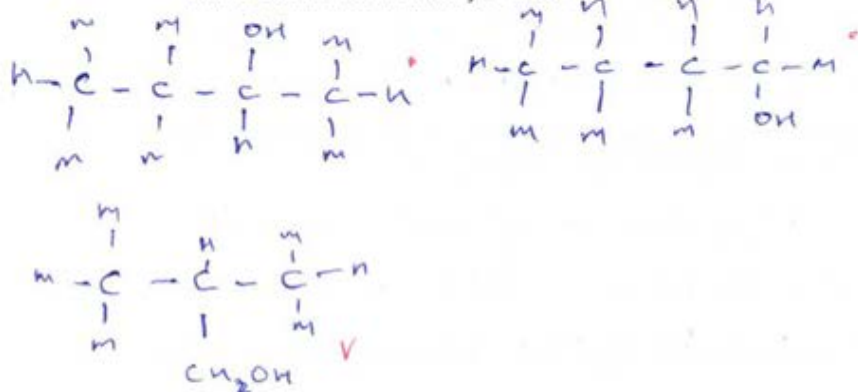
## SAMPLE CANDIDATE RESPONSE GRADED 'E'

7 Alcohols are widely used for some industrial processes.

(a) (i) An alcohol has the molecular formula  $C_4H_{10}O$ .

Draw the structure of three isomers of  $C_4H_{10}O$  that are alcohols.

Name each of the isomers you draw.



[3]

(ii) Alcohols can undergo reactions involving the loss of water.

Name this type of reaction.

elimination

[1]

(iii) State the conditions for this reaction.

strong base.

[1]

### EXAMINER'S COMMENTS

This candidate received two out of five marks. In 7a(i) the candidate received just one mark as he / she indicated the correct structural formulae of three isomers but failed to provide their names. The answer to 7a(ii) received one mark for the correct name of the process described in the task, but the candidate also lost one mark in 7a(ii) due to indicating a wrong condition for this process.

**QUESTIONS 7 b (i), (ii)**

**(b) (i)** In industry, ethanol is produced by fermentation and by the hydration of ethene.

Put ticks (✓) in the boxes to show the correct information about fermentation and hydration.

|              | higher rate of reaction | more expensive equipment | lower atom economy |
|--------------|-------------------------|--------------------------|--------------------|
| fermentation |                         |                          |                    |
| hydration    |                         |                          |                    |

[2]

(ii) Bioethanol is a biofuel made from glucose.

Use chemical equations to explain why bioethanol is a carbon-neutral fuel.

[

[3]

### MARK SCHEME

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                          |                                                                                                                                                            |                          |                    |              |  |  |   |           |   |   |  |                    |                                                                                                                                                                |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------|--------------|--|--|---|-----------|---|---|--|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7 (b) (i)    | <table><tr><td></td><td>higher rate of reaction</td><td>more expensive equipment</td><td>lower atom economy</td></tr><tr><td>fermentation</td><td></td><td></td><td>✓</td></tr><tr><td>hydration</td><td>✓</td><td>✓</td><td></td></tr></table>                                                                                                                                                                                                                                                   |                                          | higher rate of reaction                                                                                                                                    | more expensive equipment | lower atom economy | fermentation |  |  | ✓ | hydration | ✓ | ✓ |  | <div>[max 2]</div> | <div>One incorrect tick maximum 1 mark</div> <div>two incorrect ticks 0 marks</div> <div>three correct ticks 2 marks</div> <div>two correct ticks 1 mark</div> |
|              | higher rate of reaction                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | more expensive equipment                 | lower atom economy                                                                                                                                         |                          |                    |              |  |  |   |           |   |   |  |                    |                                                                                                                                                                |
| fermentation |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                          | ✓                                                                                                                                                          |                          |                    |              |  |  |   |           |   |   |  |                    |                                                                                                                                                                |
| hydration    | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ✓                                        |                                                                                                                                                            |                          |                    |              |  |  |   |           |   |   |  |                    |                                                                                                                                                                |
| 7 (b) (ii)   | <div><math display="block">6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2</math><math display="block">\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 2\text{C}_2\text{H}_5\text{OH} + 2\text{CO}_2</math><math display="block">2\text{C}_2\text{H}_5\text{OH} + 6\text{O}_2 \rightarrow 4\text{CO}_2 + 6\text{H}_2\text{O}</math></div> <div>number of moles of carbon dioxide to make glucose equals number of moles of carbon dioxide made</div> | <div>2</div> <div>1</div> <div>[3]</div> | <div>Do not accept unbalanced equation</div> <div>Note</div> <div>1 mark for one correct equation</div> <div>2 marks for all three correct equations</div> |                          |                    |              |  |  |   |           |   |   |  |                    |                                                                                                                                                                |

## SAMPLE CANDIDATE RESPONSE GRADED 'A'

- (b) (i) In industry, ethanol is produced by fermentation and by the hydration of ethene.

Put ticks (✓) in the boxes to show the correct information about fermentation and hydration.

|              | higher rate of reaction | more expensive equipment | lower atom economy |
|--------------|-------------------------|--------------------------|--------------------|
| fermentation |                         |                          | ✓                  |
| hydration    | ✓                       | ✓                        |                    |

[2]

2

- (ii) Bioethanol is a biofuel made from glucose.

Use chemical equations to explain why bioethanol is a carbon-neutral fuel.

$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$  - fermentation of the glucose ✓  
 $2C_2H_5OH + 8O_2 \rightarrow 4CO_2 + 6H_2O$  - combustion of bioethanol ✓  
 overall:  $C_6H_{12}O_6 + 8O_2 \rightarrow 8CO_2 + 6H_2O$ , which is similar ✓  
 to breathing. As a result, all of the bioethanol burns into  
 carbon dioxide which then can be used by plants for photosynthesis;  
 $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$  to produce oxygen and glucose [3]

For  
Examiner's  
Use

3

### EXAMINER'S COMMENTS

This candidate received all five marks. In 7b(i) the candidate matched all statements to corresponding processes correctly. The explanation with required balanced equations given in the answer to 7b(ii) fully complied with the mark scheme, therefore, full marks were awarded.



## SAMPLE CANDIDATE RESPONSE GRADED 'C'

- (b) (i) In industry, ethanol is produced by fermentation and by the hydration of ethene.

Put ticks (✓) in the boxes to show the correct information about fermentation and hydration.

|              | higher rate of reaction | more expensive equipment | lower atom economy |
|--------------|-------------------------|--------------------------|--------------------|
| fermentation | ✓ ✗                     |                          | ✓                  |
| hydration    |                         | ✓                        |                    |

[2]

- (ii) Bioethanol is a biofuel made from glucose.

Use chemical equations to explain why bioethanol is a carbon-neutral fuel.

~~$6H_2O + 3O_2 \rightarrow 6H_2 + 3O_2$~~   
 ~~$C_6H_{12}O_6 + O_2 \rightarrow 2C_6H_{12}O_6 \rightarrow 6C_2H_5OH + 3O_2$~~   
 $6CO_2 + 6H_2O \xrightarrow{UV} C_6H_{12}O_6 + 5O_2$  - formation of glucose requires  $CO_2$  as a reactant and produces  $O_2$ .  
 $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$  - combustion of ethanol releases the amount of  $CO_2$  equal to  $CO_2$  consumed by plants to form glucose.

For  
Examiner's  
Use

[3]

### EXAMINER'S COMMENTS

This candidate received two out of five marks. In question 7b(i) the candidate lost one mark because he / she wrongly indicated relation of the higher rate of reaction to the fermentation process. In 7b(ii) the given explanation did not include correct balanced equations as an evidence, for this reason only one mark was awarded.

## SAMPLE CANDIDATE RESPONSE GRADED 'E'

- (b) (i) In industry, ethanol is produced by fermentation and by the hydration of ethene.

Put ticks (✓) in the boxes to show the correct information about fermentation and hydration.

|              | higher rate of reaction | more expensive equipment | lower atom economy |
|--------------|-------------------------|--------------------------|--------------------|
| fermentation | ✓                       |                          |                    |
| hydration    |                         | ✓                        | ✓                  |

[2]

- (ii) Bioethanol is a biofuel made from glucose.

Use chemical equations to explain why bioethanol is a carbon-neutral fuel.

$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$  ✓  
 this  $CO_2$  will react with water  
 to produce  $C_2H_5OH$  so exchange  
 of  $CO_2$  not give opportunity to release it  
 in the atmosphere  
~~200g 137.5g 200g 137.5g~~

[3]

For  
Examiner's  
Use

### EXAMINER'S COMMENTS

This candidate received one out of five marks. In question 7b(i) the candidate received no marks because he / she wrongly indicated relation of the higher rate of reaction and the lower atom economy to the corresponding processes. In 7b(ii) the given explanation is incorrect. The candidate received only one mark for one correct balanced equation of combustion of ethanol.

### QUESTIONS 7c (i), (ii)

(c) Potassium dichromate(IV) is heated under reflux with an excess of ethanol until an obvious change in colour is detected.

(i) Describe the colour change observed during the reaction.

..... [1]

(ii) Write the organic product of this reaction.

..... [1]

### Mark scheme

|           |                                                                                   |     |                      |
|-----------|-----------------------------------------------------------------------------------|-----|----------------------|
| 7(c) (i)  | from orange to green                                                              | [1] | Accept<br>blue green |
| 7 (c)(ii) | Ethanoic acid/acetic acid/CH <sub>3</sub> COOH/ CH <sub>3</sub> CO <sub>2</sub> H | [1] |                      |

### SAMPLE CANDIDATE RESPONSE GRADED 'A'

(c) Potassium dichromate(IV) is heated under reflux with an excess of ethanol until an obvious change in colour is detected.

(i) Describe the colour change observed during the reaction.

..... [1] ✓ 1  
*from orange to green*

(ii) Write the organic product of this reaction.

..... [1] ✓ 1  
*ethanoic acid*

### EXAMINER'S COMMENTS

This candidate received full marks.

### SAMPLE CANDIDATE RESPONSE GRADED 'C'

(c) Potassium dichromate(IV) is heated under reflux with an excess of ethanol until an obvious change in colour is detected.

(i) Describe the colour change observed during the reaction.

..... From green to orange ..... X [1]

(ii) Write the organic product of this reaction.

..... ~~CH<sub>3</sub>CH<sub>2</sub>COOH~~ CH<sub>3</sub>COOH - ethanoic acid ..... ✓ [1]

0

1

#### EXAMINER'S COMMENTS

This candidate received one out of two marks. In 7c(i) the given description of the observation was incorrect.

### SAMPLE CANDIDATE RESPONSE GRADED 'E'

(c) Potassium dichromate(IV) is heated under reflux with an excess of ethanol until an obvious change in colour is detected.

(i) Describe the colour change observed during the reaction.

..... ~~from~~ from white to orange ..... X [1]

(ii) Write the organic product of this reaction.

..... 1/10/10 ..... X [1]

0

0

#### EXAMINER'S COMMENTS

This candidate received no marks. The candidate provided an incorrect description of the observation in 7c(i) and indicated an incorrect product in 7c(ii).

**QUESTIONS 8a (i), (ii), (iii)**

8 Aluminium is in Group 13 of the Periodic Table.

(a) (i) Draw a labelled diagram of the metallic structure of aluminium.

[3]

(ii) Explain why aluminium has a high melting point.

.....

.....

.....

.....

..... [2]

(iii) Explain why aluminium has a high thermal conductivity.

.....

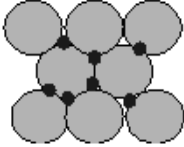
.....

.....

.....

..... [2]

## Mark scheme

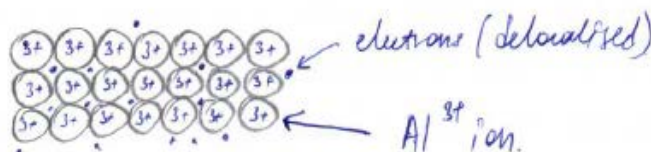
|            |                                                                                                                                               |                                              |                                                                                                           |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 8(a)(i)    | <p>Correct structure</p> <p>Aluminum ions</p> <p>Free electrons/delocalized electrons/sea of electrons</p>                                    | <p>1</p> <p>1</p> <p>1</p> <p><b>[3]</b></p> | <p>Correct diagram</p>  |
| 8(a) (ii)  | <p>It has crystal/metallic lattice/metallic bond</p> <p>Strong electrostatic attraction / strong attraction between cations and electrons</p> | <p>1</p> <p>1</p> <p><b>[2]</b></p>          | <p><b>Accept</b><br/> <math>Al^{3+}</math> <math>Al^{2+}</math> <math>Al^+</math> instead of cation</p>   |
| 8(a) (iii) | <p>heat transferred</p> <p>by mobile electrons/ kinetic energy of electrons</p>                                                               | <p>1</p> <p>1</p> <p><b>[2]</b></p>          |                                                                                                           |



## SAMPLE CANDIDATE RESPONSE GRADED 'A'

8 Aluminium is in Group 13 of the Periodic Table.

(a) (i) Draw a labelled diagram of the metallic structure of aluminium.



Metallc structure of Aluminium.

[3]

(ii) Explain why aluminium has a high melting point.

Aluminium has a giant metallic structure. There is a strong electrostatic attraction between positively charged ion ( $Al^{3+}$ ) and the 'sea' of delocalised electrons. Thus to overcome this electrostatic force, a lot of energy required  $\Rightarrow$  high melting point.

(iii) Explain why aluminium has a high thermal conductivity.

Each  $Al^{3+}$  ion releases 3 electrons, so there is a huge 'sea' of delocalised electrons in the lattice. These delocalised electrons are carriers of heat/energy. Heat is also transferred by vibrations of ~~interpositive~~ nuclei.  $\Rightarrow$  Al has a high thermal conductivity as it has a lot of mobile electrons.

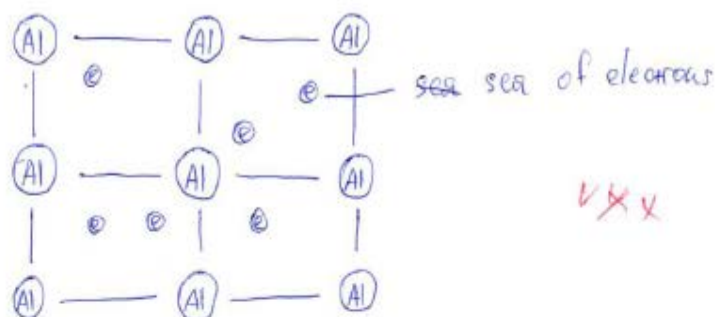
### EXAMINER'S COMMENTS

This candidate received all seven marks. The given diagram in 8a (i) was drawn correctly with all species labelled and therefore received full marks. The answers in 8a(ii) and 8a(iii) included a complete explanation of metallic properties required in the task using key terms as suggested by the mark scheme.

## SAMPLE CANDIDATE RESPONSE GRADED 'C'

8 Aluminium is in Group 13 of the Periodic Table.

(a) (i) Draw a labelled diagram of the metallic structure of aluminium.



(ii) Explain why aluminium has a high melting point.

because it has a giant metallic structure and small radius. Thus, it is difficult to break this structure. Thus, because of sea of electrons and short bonds, it is difficult to break the bonding. Also aluminium has only one electron on 3p orbital, thus therefore a lot of energy is required to lose it. [2]

(iii) Explain why aluminium has a high thermal conductivity.

because of & transfer sea of electrons.

[2]

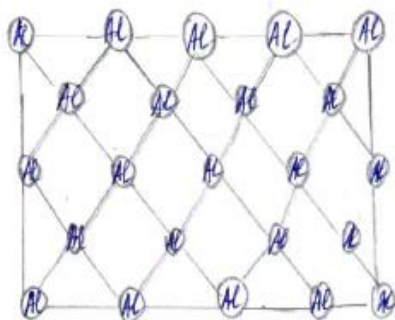
### EXAMINER'S COMMENTS

This candidate received three out of seven marks. He / she lost two marks in question 8a(i) due to drawing an incorrect diagram in point of rods that connect metal atoms together. In 8a(ii) one mark was awarded for an incomplete explanation of high melting point. The candidate determined correctly the type of lattice which directly influenced the considered physical property but failed to explain clearly the reason why this structure effects in such way. In part 8a(iii) the candidate received one mark instead of two due to partial explanation which included only the identification of species responsible for the high thermal conductivity.

## SAMPLE CANDIDATE RESPONSE GRADED 'E'

8 Aluminium is in Group 13 of the Periodic Table.

(a) (i) Draw a labelled diagram of the metallic structure of aluminium.



[3]

(ii) Explain why aluminium has a high melting point.

Al is metal, so in Metals high melting point. Aluminium also has a high melting point, because Al's type of bond is Metallic. Metallic bond too hard to break. So a lot of energy needed to break. So melting point is high.

(iii) Explain why aluminium has a high thermal conductivity.

Al is metal, Metals are good conductors. Al has metallic bond which is hard to break down, so it has a high thermal conductivity.

[2]

### EXAMINER'S COMMENTS

This candidate received one out of seven marks.

### QUESTION 8b

(b) Write an equation to represent the third ionisation energy of aluminium.

Include state symbols in your equation.

..... [2]

### MARK SCHEME

|       |                                                                                |     |                                                                                                                            |
|-------|--------------------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------------------------------------|
| 8 (b) | $\text{Al}^{2+}(\text{g}) \rightarrow \text{Al}^{3+}(\text{g}) + \text{e}^{-}$ | [2] | <b>Note</b><br><br>1 mark for correct equation<br><br>1 mark for correct state symbols – dependent on correct formulae (1) |
|-------|--------------------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------------------------------------|

### SAMPLE CANDIDATE RESPONSE GRADED ‘A’

(b) Write an equation to represent the third ionisation energy of aluminium.

Include state symbols in your equation.

$\text{Al}^{2+}(\text{g}) \rightarrow \text{Al}^{3+}(\text{g}) + \text{e}^{-}$  ..... [2] ✓✓

2

#### EXAMINER’S COMMENTS

This candidate received full marks for the correct equation and states of included species.

### SAMPLE CANDIDATE RESPONSE GRADED ‘C’

(b) Write an equation to represent the third ionisation energy of aluminium.

Include state symbols in your equation.

$\text{Al}^{2+} \rightarrow \text{Al}^{3+} + \text{e}^{-}$  ..... [2] ✓x

1

#### EXAMINER’S COMMENTS

This candidate received one out of two marks for the correct equation without states of included species.

## SAMPLE CANDIDATE RESPONSE GRADED 'E'

- (b) Write an equation to represent the third ionisation energy of aluminium.

Include state symbols in your equation.



### EXAMINER'S COMMENTS

This candidate received no marks due to incorrect equation and states of included species.

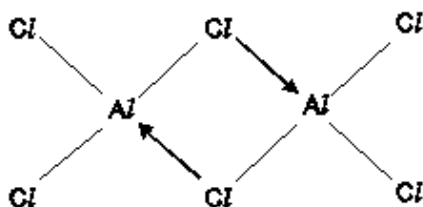
## QUESTIONS 8 c, d

- (c) Draw the 'dot-and-cross' diagram for a molecule of aluminium chloride,  $\text{AlCl}_3$ .

Include only the outer shell electrons.

[2]

- (d) At low temperatures aluminium chloride forms a dimer with the formula  $\text{Al}_2\text{Cl}_6$ .



Suggest why  $\text{AlCl}_3$  molecules form the dimer  $\text{Al}_2\text{Cl}_6$ .

.....  
.....  
.....  
.....  
..... [2]

## MARK SCHEME

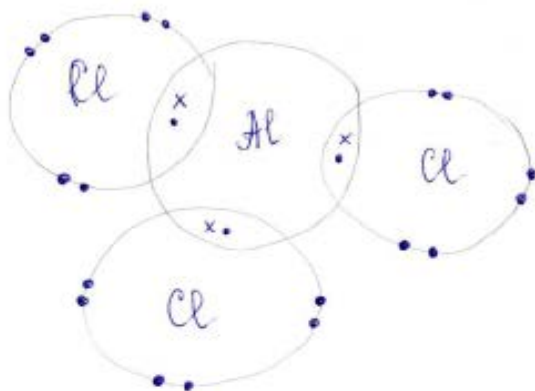
|       |                                                                                                                                                                                                                                                 |                                     |                                                                                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8 (c) | <p>Dot-and-cross diagram showing:</p> <p>Lack/deficiency of electrons on the outer shell of <u>aluminium</u></p> <p>Lone pair from two chlorine atoms forms dative/coordinate bonding with <u>aluminium</u> atoms</p>                           | <p>1</p> <p>1</p> <p><b>[2]</b></p> |                                                                                                                                                                 |
| 8 (d) | <p><u>Aluminium</u> requires 5 electrons to complete its octet.</p> <p>Monomer (<math>\text{AlCl}_3</math>) is unstable due to unpaired electron in p-orbital of Al, dimer (<math>\text{Al}_2\text{Cl}_6</math>) so that it becomes stable.</p> | <p>1</p> <p>1</p> <p><b>[2]</b></p> | <p><b>Accept</b></p> <p>Dimer forms through a coordinate/dative covalent bond to complete the octet of <u>aluminium</u> using lone pairs of chlorine atoms.</p> |



## SAMPLE CANDIDATE RESPONSE GRADED 'A'

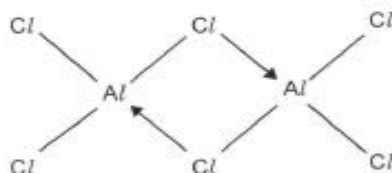
- (c) Draw the 'dot-and-cross' diagram for a molecule of aluminium chloride,  $AlCl_3$ .

Include only the outer shell electrons.



[2]

- (d) At low temperatures aluminium chloride forms a dimer with the formula  $Al_2Cl_6$ .



Suggest why  $AlCl_3$  molecules form the dimer  $Al_2Cl_6$ .

$AlCl_3$  is electron-deficient, so in order to reach stability, full octet (8 electrons on the outermost shell) accordingly to the octet rule,  $Al_2Cl_6$  forms with dative bonds, where Cl donates electron pair to Al.

[2]

### EXAMINER'S COMMENTS

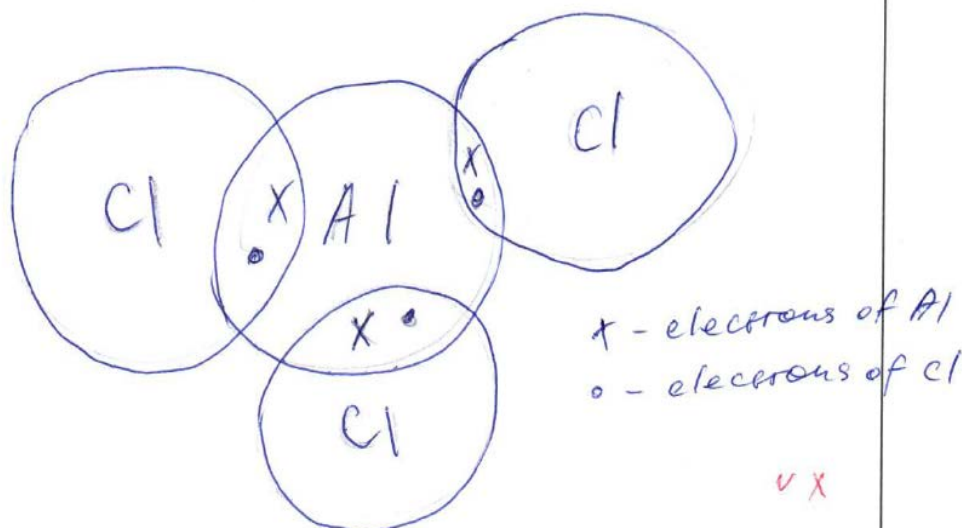
This candidate received all four marks. In part 8c the drawn 'dot-and-cross' diagram received both marks for correct bond pairs and lone pairs in the molecule. In question 8d the given answer was full in point of the idea about complete octet and stabilization of the  $Al_2Cl_6$ .

# **SAMPLE CANDIDATE RESPONSE GRADED 'C'**

(c) Draw the 'dot-and-cross' diagram for a molecule of aluminium chloride,  $\text{AlCl}_3$ .

Include only the outer shell electrons.

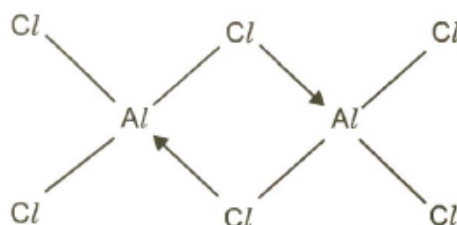
For  
Examiner's  
Use



v x

[2]

(d) At low temperatures aluminium chloride forms a dimer with the formula  $\text{Al}_2\text{Cl}_6$ .



Suggest why  $\text{AlCl}_3$  molecules form the dimer  $\text{Al}_2\text{Cl}_6$ .

Cl has 3 lone pairs, & thus it donates both electrons to 1st Al to form dative bond with it. ✓ Cl on the 2nd Al donates pair of electrons to 1st Al to form dative bond. x

[2]

1

### EXAMINER'S COMMENTS

This candidate received two out of four marks. In 8c the drawn 'dot-and-cross' diagram received one mark for correct bond pairs and lost one due to absence of lone pairs in the molecule. The answer to question 8d received only one mark due to the explanation of formation of dative covalent bonds without mentioning the idea about stability of dimer.

### SAMPLE CANDIDATE RESPONSE GRADED 'E'

(c) Draw the 'dot-and-cross' diagram for a molecule of aluminium chloride,  $\text{AlCl}_3$ .

Include only the outer shell electrons.

For  
Examiner's  
Use

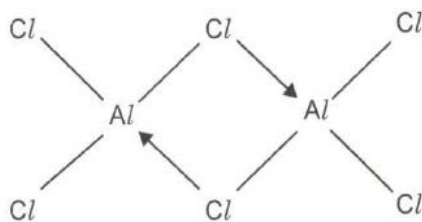


X  
X

[2]

0

(d) At low temperatures aluminium chloride forms a dimer with the formula  $\text{Al}_2\text{Cl}_6$ .



Suggest why  $\text{AlCl}_3$  molecules form the dimer  $\text{Al}_2\text{Cl}_6$ .

*$\text{AlCl}_3$  has a empty subshell space to accept the electrons. X*  
 *$\text{AlCl}_3$  bind  $\text{Cl}$  forming  $[\text{AlCl}_4]^-$*   
*then at low temperatures  $\text{AlCl}_3$  with empty space take electrons from  $[\text{AlCl}_4]^-$  X [2]*  
*forming the dimer.*

#### EXAMINER'S COMMENTS

This candidate received no marks. In 8c the 'dot-and-cross' diagram for  $\text{AlCl}_3$  was incorrect. In question 8d the given answer was incorrect in point of the idea about complete octet and stabilization of the  $\text{Al}_2\text{Cl}_6$ .